Disease: Common Cold (Viral upper respiratory infection)

Overview:

- Usually caused by rhinoviruses, coronaviruses, or other respiratory viruses.
- Mild, self-limited infection of nose/throat lasting ~3–10 days.
- Transmission via respiratory droplets and contaminated surfaces.
- Most people recover without complications; infants, elderly, chronic disease at higher risk of complications.
- Complications: secondary bacterial sinusitis/otitis media or asthma exacerbation.
- Seek care for high fever, severe breathing difficulty, or symptoms that worsen after initial improvement.

Common Symptoms:

- Nasal congestion, runny nose
- Sneezing, sore throat
- Mild cough, hoarseness
- Low-grade fever (often absent in adults)
- Malaise, mild headache

Precautions (after suspicion/diagnosis):

- Rest, fluids, symptomatic care; stay home when contagious.
- Hand hygiene, tissues for sneezes, mask if close contact with vulnerable people.
- Avoid antibiotics unless bacterial infection suspected.
- Monitor for high fever, difficulty breathing, worsening symptoms.

Prevention:

- Hand washing, respiratory etiquette, avoid close contact with sick people.
- Keep surfaces clean; avoid touching face.
- Flu and COVID vaccination as appropriate to reduce similar respiratory illnesses.

Medications_and_Dosage (general reference, not a prescription):

- Paracetamol/acetaminophen:
    - Adults: 500–1000 mg every 4–6 h as needed (max 3–4 g/day; prefer ≤3 g/day in many guidelines).
    - Children: 10–15 mg/kg every 4–6 h (per label; max per 24 h by weight).
    - Older adults: use lower max doses if liver disease.
- Ibuprofen (if no contraindication):
    - Adults: 200–400 mg every 6–8 h as needed.

    - Children: 5–10 mg/kg every 6–8 h (per label).
- Nasal saline spray/drops: all ages — supportive.
- Oral decongestants (pseudoephedrine) and topical nasal decongestant sprays (oxymetazoline) — adults only or per label; avoid prolonged use of topical sprays >3 days (rebound congestion); avoid sympathomimetics in uncontrolled hypertension.

#END#

Disease: Influenza (Flu)

Overview:

- Acute respiratory illness caused mainly by influenza A and B viruses.
- Rapid onset of fever, systemic symptoms, and respiratory signs; duration usually 5–7 days.
- Can cause severe disease in elderly, young children, pregnant people, and those with comorbidities.
- Complications: viral or secondary bacterial pneumonia, myocarditis, encephalitis, worsening of chronic diseases.
- Annual epidemics; vaccination updated yearly.
- Seek urgent care for severe respiratory distress, chest pain, decreased responsiveness, or very high/prolonged fever.

Common Symptoms:

- Sudden fever and chills
- Cough, sore throat, nasal congestion
- Myalgia, headache, severe fatigue
- Sometimes vomiting/diarrhoea (more in children)

Precautions (after suspicion/diagnosis):

- Isolate at home, rest, fluids, mask around vulnerable people.
- Antiviral treatment if high-risk or severe — start early (ideally within 48 hours) as per clinician.
- Avoid aspirin in children and teens.

Prevention:

- Annual influenza vaccination.
- Hand hygiene, respiratory etiquette, avoid close contact during outbreaks.

Medications_and_Dosage (general reference, not a prescription):

- Oseltamivir (oral neuraminidase inhibitor):
    - Adults: 75 mg twice daily for 5 days (treatment); prophylaxis regimens differ.

    - Children: weight-based dosing per label (e.g., ~3 mg/kg twice daily for some age groups); follow product insert.
    - Older adults: same dosing; adjust if renal impairment.
    - Notes: best if started within 48 hours; GI side effects possible.
- Supportive: acetaminophen/NSAIDs per doses above.

#END#

Disease: COVID-19 (SARS-CoV-2 Infection)

Overview:

- Viral infection caused by SARS-CoV-2 with a wide spectrum from asymptomatic to severe respiratory failure.
- Transmission mainly via respiratory droplets/aerosols; incubation typically 2–7 days.
- Acute disease may progress to hypoxemic pneumonia, thrombotic complications, multisystem inflammatory syndromes.
- Long COVID (post-acute sequelae) can cause persistent fatigue, cognitive symptoms, dyspnoea.
- Vaccination and early outpatient antivirals for high-risk patients reduce progression.
- Seek urgent care for severe breathlessness, persistent chest pain, syncope, altered mental status, or low oxygen saturation.

Common Symptoms:

- Fever, cough, sore throat
- Shortness of breath, loss of taste/smell
- Fatigue, myalgias, headache
- GI symptoms in some cases (nausea, diarrhoea)

Precautions (after suspicion/diagnosis):

- Isolate per local guidance; wear mask around others if present.
- Monitor symptoms and oxygen saturation at home if advised; seek care for worsening.
- Follow clinician guidance about antivirals, monoclonal antibodies (where available/indicated), and supportive care.

Prevention:

- Vaccination and boosters as recommended.
- Masking, ventilation, hand hygiene; testing when symptomatic or exposed.

Medications_and_Dosage (general reference, not a prescription):

- Nirmatrelvir/ritonavir (Paxlovid) — outpatient treatment for eligible high-risk adults:

    - Adults: nirmatrelvir 300 mg (two 150 mg tablets) with ritonavir 100 mg twice daily for 5 days (adjustments if eGFR <60 mL/min; drug–drug interactions common).
    - Children: not routinely for children under specified age/weight; follow local paediatric guidance.
    - Notes: multiple drug interactions due to ritonavir; check interactions before use.
- Remdesivir (IV) — for hospitalized patients as per hospital protocol.
- Dexamethasone 6 mg IV/PO daily up to 10 days — for patients requiring supplemental oxygen.
- Antipyretics and supportive care per ages above.

#END#

Disease: Dengue Fever (Vector-borne Flavivirus)

Overview:

- Mosquito-borne infection (Aedes spp.) caused by dengue viruses (4 serotypes).
- Incubation 4–10 days; abrupt high fever with severe myalgia/arthralgia and headache common.
- Some progress to severe dengue with plasma leakage, bleeding, shock; critical phase often around day 3–7.
- No specific antiviral; management is mainly careful supportive fluid management and monitoring.
- Prior infection with different serotype increases severe disease risk (antibody-dependent enhancement).
- Seek urgent care for severe abdominal pain, persistent vomiting, bleeding, rapid pulse with low BP, lethargy.

Common Symptoms:

- High fever, severe headache (often retro-orbital)
- Severe muscle/joint pain, rash
- Mild bleeding (gums/nose), easy bruising
- Nausea, vomiting

Precautions (after suspicion/diagnosis):

- Maintain oral hydration; hospitalize and give IV fluids for warning signs or inability to tolerate oral intake.
- Avoid NSAIDs and aspirin (bleeding risk); use acetaminophen for fever/pain.
- Monitor haematocrit and platelet counts if advised.

Prevention:

- Mosquito bite prevention (repellents, nets, clothing).
- Eliminate standing water and community vector control.
- Vaccination policies vary by country and require prior infection testing in some settings.

Medications_and_Dosage (general reference, not a prescription):

- Acetaminophen:
  - Adults: 500–1000 mg every 4–6 h as needed (max per local guidance; avoid excessive total daily dose).
  - Children: 10–15 mg/kg every 4–6 h per label.
- Oral rehydration solutions or IV fluids as clinically indicated (no single universal dose; fluid management guided by clinical status and haematocrit).

#END#

Disease: Malaria

Overview:

- Protozoal infection transmitted by Anopheles mosquitoes; species include Plasmodium falciparum, vivax, ovale, malariae, knowlesi.
- Presents with cyclical fevers, chills, sweats; severity ranges from uncomplicated to severe/fatal (especially P. falciparum).
- Diagnosis via rapid diagnostic test or blood smear; prompt species-directed treatment required.
- Prevention includes mosquito control, chemoprophylaxis for travellers to endemic areas, and prompt treatment to prevent complications.
- Severe malaria is a medical emergency (cerebral malaria, severe anaemia, renal failure).

Common Symptoms:

- Fever with rigors/chills, sweats
- Headache, myalgia, malaise
- Nausea, vomiting; sometimes cough or diarrhoea
- Jaundice, dark urine in severe disease

Precautions (after suspicion/diagnosis):

- Seek prompt diagnostic testing and start appropriate antimalarial therapy as directed.
- Monitor for signs of severe disease (confusion, respiratory distress, hypotension, severe anaemia).

- Avoid self-medicating without confirmed diagnosis and species identification when possible.

Prevention:

- Mosquito bite prevention: bed nets, repellents, indoor spraying.
- Antimalarial chemoprophylaxis for travellers (drug choice depends on region/resistance).
- Prompt diagnosis and treatment reduce transmission.

Medications_and_Dosage (general reference, not a prescription):

- Uncomplicated P. falciparum (example: artemisinin-based combination therapy — ACT such as artemether-lumefantrine):
    - Adults: artemether-lumefantrine fixed-dose twice daily for 3 days (dosing per product; commonly 20/120 mg per tablet, 4 tablets per dose initially then 4 tablets at 8, 24, 36, 48, 60 hours — follow label).
    - Children: weight-based dosing per product insert.
    - Notes: follow country-specific first-line ACTs; adherence important.
- Severe malaria:
    - IV artesunate (hospital only) — dosing per WHO/hospital protocol (e.g., 2.4 mg/kg IV at 0, 12, 24 h then daily) then oral ACT when able; management in ICU may be required.
- Primaquine (for radical cure of P. vivax/ovale hypnozoites):
    - Adults: e.g., 15 mg base daily for 14 days (varies by regimen) after ruling out G6PD deficiency.
    - Children: weight-based; check G6PD prior to use.
    - Notes: G6PD testing essential due to haemolysis risk.

#END#

Disease: Typhoid (Enteric Fever)

Overview:

- Systemic infection usually caused by Salmonella Typhi (and Paratyphi).
- Transmission faecal–oral via contaminated food/water; incubation ~1–3 weeks.
- Presents with prolonged fever, abdominal discomfort, constipation or diarrhoea, and sometimes rose spots.
- Complications: intestinal perforation, haemorrhage, sepsis if untreated.
- Diagnosis by blood cultures; treat with antibiotics guided by susceptibility.
- Seek urgent care for severe abdominal pain, high fever, confusion, GI bleeding.

Common Symptoms:

- Prolonged high fever, headache
- Abdominal pain, constipation or diarrhoea
- Malaise, anorexia; sometimes cough early on
- Relative bradycardia and rose-coloured maculopapular rash in some

Precautions (after suspicion/diagnosis):

- Start appropriate antibiotic therapy per local susceptibility patterns; ensure hydration.
- Practice strict hygiene to avoid transmission (handwashing, safe food handling).
- Monitor for GI complications and signs of systemic spread.

Prevention:

- Vaccination available (oral/live-attenuated and injectable Vi polysaccharide) for travellers and at-risk populations.
- Safe water, sanitary food handling, hand hygiene.

Medications_and_Dosage (general reference, not a prescription):

- Empiric antibiotics vary with resistance patterns. Examples where susceptible:
    - Azithromycin (oral):
        - Adults: 500 mg once daily for 5–7 days.
        - Children: 10 mg/kg once daily (max adult dose) for 5–7 days.
        - Notes: used in areas with fluoroquinolone resistance.
    - Ceftriaxone (IV) for severe disease:
        - Adults: 1–2 g IV once daily.
        - Children: 50–100 mg/kg IV once daily (per hospital protocols).
    - Ciprofloxacin (where susceptible) — dosing per label; many regions have resistance.
- Supportive care: fluids, antipyretics per ages above.

#END#

Disease: Hepatitis A (Acute Viral Hepatitis, faecal-oral)

Overview:

- Acute self-limited liver infection caused by hepatitis A virus (HAV).
- Transmitted faecal–oral through contaminated food/water or close contact.
- Incubation ~2–6 weeks; symptomatic disease more common in adults.

- Most recover fully; fulminant hepatic failure is rare but higher risk in older adults and those with chronic liver disease.
- Prevention via vaccine and improved sanitation.
- Seek urgent care for jaundice with confusion, bleeding, or signs of liver failure.

Common Symptoms:

- Fever, malaise, anorexia
- Nausea, vomiting, abdominal discomfort (RUQ)
- Jaundice, dark urine, pale stools
- Itching (cholestatic symptoms) in some

Precautions (after suspicion/diagnosis):

- Supportive care, avoid alcohol and hepatotoxic drugs.
- Maintain hydration and nutrition; limit strenuous activity during acute phase.
- Household contacts may need post-exposure prophylaxis (vaccine or immunoglobulin) per local guidance.

Prevention:

- Hepatitis A vaccination (routine or travel-related).
- Safe water, hand hygiene, sanitation, safe food practices.

Medications_and_Dosage (general reference, not a prescription):

- No specific antiviral; supportive care only.
- Symptom control:
    - Paracetamol: use cautiously and at lower total daily dose due to liver involvement; consult clinician.
- Post-exposure prophylaxis:
    - Hepatitis A vaccine or immune globulin dosing per public health guidance (age- and risk-dependent).

#END#

Disease: Hepatitis B (HBV Infection)

Overview:

- Viral infection affecting the liver; can be acute or progress to chronic infection.
- Transmitted via blood, sexual contact, or perinatal exposure.
- Chronic HBV increases risk of cirrhosis and hepatocellular carcinoma.
- Many acute infections are asymptomatic; chronic infection often silent until advanced.

- Screening and antiviral therapy can reduce complications; vaccination prevents infection.
- Seek urgent care for jaundice with encephalopathy, bleeding, or signs of acute liver failure.

Common Symptoms:

- Fatigue, anorexia, nausea
- Right upper quadrant discomfort, jaundice
- Dark urine, pale stools
- Many chronic cases asymptomatic

Precautions (after suspicion/diagnosis):

- Counsel on preventing transmission (safe sex, avoid sharing needles).
- Screen household contacts and consider vaccination.
- Monitor liver function tests and viral markers; refer to specialist for antiviral therapy if criteria met.

Prevention:

- Hepatitis B vaccination (infant series and catch-up for unvaccinated).
- Safe injection practices, blood screening, safe sex, antenatal screening and peripartum prophylaxis.

Medications_and_Dosage (general reference, not a prescription):

- Antiviral agents for chronic HBV (selected examples):
    - Tenofovir disoproxil fumarate (TDF):
        - Adults: 300 mg orally once daily.
        - Children: weight-based formulations per label.
        - Notes: monitor renal function and bone health.
    - Entecavir:
        - Adults: 0.5 mg once daily (1 mg if lamivudine-resistant).
        - Children: weight-based dosing; specialist-guided.
    - Treatment indications depend on viral load, ALT, fibrosis stage; managed by specialists.

#END#

Disease: Hepatitis C (HCV Infection)

Overview:

- Bloodborne viral infection that often becomes chronic and can lead to cirrhosis and hepatocellular carcinoma.
- Transmission mainly via infected blood; less commonly sexual or perinatal.
- Many are asymptomatic until advanced liver disease; now curable with direct-acting antivirals (DAAs).
- Genotype used historically to guide therapy, though many regimens are pan-genotypic.
- Screening is important to identify candidates for curative therapy.
- Seek specialist care for staging and treatment; urgent care for acute liver failure signs.

Common Symptoms:

- Often asymptomatic in chronic infection
- Fatigue, mild abdominal discomfort
- Jaundice typically only in advanced disease
- Extrahepatic manifestations (e.g., cryoglobulinemia) in some

Precautions (after suspicion/diagnosis):

- Avoid alcohol and hepatotoxic medications.
- Counsel to prevent transmission (no sharing needles, safe sex as advised).
- Baseline and periodic monitoring of liver function and viral RNA; refer for DAA therapy.

Prevention:

- No vaccine currently; harm-reduction strategies (needle programs), blood screening, safe medical practices.

Medications_and_Dosage (general reference, not a prescription):

- Direct-acting antivirals (examples of common pan-genotypic regimens — dosing per product label and specialist guidance):
    - Sofosbuvir/velpatasvir:
        - Adults: one fixed-dose tablet once daily for 12 weeks (typical adult dosing).
        - Children: paediatric formulations and dosing vary by age/weight; specialist-guided.
        - Notes: high cure rates; check drug interactions and renal function for some regimens.
- Treatment chosen and duration individualized; management by hepatology/infectious disease specialists.

#END#

Disease: Tuberculosis (Pulmonary TB)

Overview:

- Infection by Mycobacterium tuberculosis affecting mainly the lungs but can be extrapulmonary.
- Transmission via airborne droplet nuclei; requires prolonged close exposure.
- Active TB presents with cough, weight loss, night sweats, fever; latent TB is asymptomatic.
- Diagnosis via sputum smear/culture, molecular tests, chest imaging.
- Treatment requires prolonged multi-drug therapy; adherence and resistance testing critical.
- Seek urgent care for haemoptysis, severe breathlessness, or if TB symptoms persist.

Common Symptoms:

- Persistent cough (>2 weeks), sometimes haemoptysis
- Fever (low-grade), night sweats, weight loss
- Fatigue, loss of appetite, chest pain

Precautions (after suspicion/diagnosis):

- Airborne precautions (masking, isolation) until non-infectious per local guidelines.
- Start appropriate multi-drug anti-TB therapy based on susceptibility.
- Ensure adherence, contact tracing, and screening of close contacts.

Prevention:

- Bacille Calmette–Guérin (BCG) vaccination policies vary (often given to infants in high-burden countries).
- Early detection and treatment of active cases reduce transmission.
- Good ventilation and respiratory precautions in healthcare settings.

Medications_and_Dosage (general reference, not a prescription):

- First-line regimen for drug-sensitive TB (initial intensive phase example):
    - Isoniazid (INH), Rifampicin (Rifampin), Pyrazinamide, Ethambutol (RIPE):
        - Adults: doses per weight — e.g., Isoniazid 300 mg once daily, Rifampicin 600 mg once daily, Pyrazinamide ~20–25 mg/kg once daily, Ethambutol ~15–20 mg/kg once daily (exact dosing per weight/tables).
        - Children: weight-based dosing per paediatric TB guidelines.
        - Older adults: adjust for weight, liver function; monitor for hepatotoxicity and drug interactions.

    - Continuation phase often INH + Rifampicin for 4 months (standard total 6 months for drug-sensitive TB) — follow guidelines.
- Pyridoxine (vitamin B6) often given with INH (e.g., 25–50 mg daily) to prevent neuropathy, especially in malnourished or pregnant patients.
- Drug-resistant TB regimens are complex and specialist-managed.

#END#

Disease: Bacterial Pneumonia (Community-Acquired)

Overview:

- Infection of lung parenchyma commonly due to Streptococcus pneumoniae, H. influenzae, atypicals (Mycoplasma), or others.
- Presents with cough, fever, sputum production, and radiographic consolidation.
- Severity ranges from outpatient treatable to life-threatening requiring hospitalization.
- Diagnosis by clinical exam, chest X-ray, sometimes sputum culture.
- Seek urgent care for breathlessness, hypoxia, sepsis signs, or inability to tolerate oral intake.

Common Symptoms:

- Fever, productive cough
- Pleuritic chest pain, shortness of breath
- Fatigue, tachypnoea, sometimes confusion in elderly

Precautions (after suspicion/diagnosis):

- Start appropriate empiric antibiotics based on severity/local guidelines.
- Supportive care, oxygen if hypoxic, hydration, and follow-up chest imaging as indicated.
- Monitor for sepsis or respiratory failure.

Prevention:

- Vaccination: pneumococcal vaccines per age/ risk guidelines; influenza vaccination reduces secondary pneumonia.
- Smoking cessation, good chronic disease management.

Medications_and_Dosage (general reference, not a prescription):

- Outpatient typical first-line (examples; modify per local resistance and comorbidity):
    - Amoxicillin:
        - Adults: 500 mg three times daily or 875 mg twice daily for typical bacterial coverage.
        - Children: weight-based dosing (e.g., 40–90 mg/kg/day divided).

    - Doxycycline or a macrolide (azithromycin) depending on atypical coverage and local resistance:
        - Azithromycin adults: 500 mg day 1 then 250 mg daily days 2–5 (or 500 mg once daily for 3 days).
- Inpatient/severe disease:
    - Ceftriaxone 1–2 g IV daily plus azithromycin 500 mg IV/PO daily, or respiratory fluoroquinolone monotherapy (where indicated) — follow local guidelines.
- Oxygen therapy, fluids, and supportive care as needed.

#END#

Disease: Acute Viral Gastroenteritis (Norovirus / Rotavirus)

Overview:

- Sudden-onset infection of the GI tract causing vomiting and diarrhea; rotavirus common in young children, norovirus common in all ages and outbreaks.
- Usually self-limited (24–72 hours) but can cause dehydration, especially in infants and elderly.
- Transmission via faecal–oral route and contaminated food/surfaces; highly contagious.
- Management is supportive with rehydration and electrolyte replacement.
- Seek care for signs of dehydration, high fever with bloody stool, or prolonged symptoms.

Common Symptoms:

- Nausea, vomiting, watery diarrhoea
- Abdominal cramps, low-grade fever
- Dehydration signs (dry mucous membranes, decreased urine output)

Precautions (after suspicion/diagnosis):

- Maintain hydration with oral rehydration solutions; hospitalize if severe dehydration.
- Strict hand hygiene and environmental cleaning to prevent spread.
- Isolate from food preparation and vulnerable persons until recovery.

Prevention:

- Rotavirus vaccination for infants as routine in many countries.
- Hand washing, safe food handling, surface disinfection; avoid preparing food while ill.

Medications_and_Dosage (general reference, not a prescription):

- Oral Rehydration Solution:

    - Children and adults: small frequent sips; volumes based on degree of dehydration and age — follow WHO/ local guidelines.
- Antiemetics (e.g., ondansetron) in children to control vomiting under clinician guidance:
    - Children (oral disintegrating): typical single dose 2–8 mg depending on age/weight; clinician-prescribed.
- Avoid antimotility agents in children and if bloody diarrhoea or high fever (possible invasive bacterial infection).

#END#

Disease: Cholera

Overview:

- Acute diarrheal illness caused by Vibrio cholerae, producing profuse watery “rice-water” stools and rapid dehydration.
- Transmitted through contaminated water/food; outbreaks in areas with poor sanitation.
- Prompt rehydration is lifesaving; antibiotics reduce duration and volume of diarrhoea in moderate-severe cases.
- Seek urgent care for profuse watery diarrhoea, signs of severe dehydration, hypotension, or altered mental status.

Common Symptoms:

- Sudden, profuse watery diarrhoea
- Vomiting, rapid dehydration
- Muscle cramps from electrolyte loss

Precautions (after suspicion/diagnosis):

- Immediate rehydration (ORS for mild/moderate, IV isotonic fluids for severe dehydration).
- Monitor electrolytes, urine output, and vital signs.
- Isolate and practice hygiene to prevent spread.

Prevention:

- Safe water, sanitation, food hygiene, and cholera vaccine in at-risk areas per public health guidance.

Medications_and_Dosage (general reference, not a prescription):

- Rehydration (primary treatment):
    - Oral Rehydration Solution per WHO formulas; IV isotonic crystalloids (e.g., Ringer’s lactate or normal saline) for severe dehydration — volumes based on severity and weight.

- Antibiotics (short-course to reduce stool volume and duration in moderate-severe cases; choose per sensitivity):
    - Doxycycline single dose (adults): e.g., 300 mg single dose in some settings (varies); children and pregnant women get alternatives (e.g., azithromycin).
    - Azithromycin children: weight-based single or short-course dosing per local guidelines.
- Zinc supplementation in children (reduces duration of diarrhoea):
    - Children: 10–20 mg daily for 10–14 days (age-dependent).

#END#

Disease: Typhus & Rickettsial Infections (including Scrub Typhus)

Overview:

- Caused by Rickettsia or Orientia species; scrub typhus (Orientia tsutsugamushi) transmitted by chigger bites, others by lice/mites/ticks.
- Presents with fever, headache, sometimes rash, and an eschar at bite site (scrub typhus).
- Can lead to multi-organ involvement if untreated.
- Diagnosed clinically and confirmed by serology or molecular tests where available.
- Seek urgent care for rapidly worsening disease, respiratory distress, neurological signs, or organ dysfunction.

Common Symptoms:

- High fever, severe headache
- Maculopapular rash in some; eschar (black scab) in scrub typhus
- Myalgia, cough, GI symptoms

Precautions (after suspicion/diagnosis):

- Start appropriate antibiotics promptly when clinical suspicion is high (empiric therapy often advised).
- Supportive care and monitor for complications (renal, hepatic, respiratory involvement).

Prevention:

- Avoid mite/tick exposure, use repellents, protective clothing.
- Early recognition in endemic areas; control vectors.

Medications_and_Dosage (general reference, not a prescription):

- Doxycycline (treatment of choice for many rickettsial infections):
    - Adults: 100 mg twice daily for 5–7 days (duration may vary).

    - Children: doxycycline may be used for severe scrub typhus even in children (short courses) — specialist guidance recommended; traditionally tetracyclines avoided in young children but doxycycline short-course often recommended in severe rickettsial disease; follow local paediatric guidance.
    - Older adults: dose as adult; monitor for photosensitivity and GI effects.
- Azithromycin is an alternative in pregnancy or where doxycycline contraindicated:
    - Adults: 500 mg once daily for 3 days or 500 mg day 1 then 250 mg days 2–5 depending on regimen and organism.

#END#

Disease: Leptospirosis

Overview:

- Zoonotic bacterial infection (Leptospira spp.) transmitted via contact with contaminated water/soil or animal urine.
- Clinical spectrum ranges from mild flu-like illness to severe Weil's disease (jaundice, renal failure, haemorrhage).
- Early diagnosis and appropriate antibiotics improve outcomes.
- Seek urgent care for jaundice, oliguria, haemorrhage, severe respiratory distress, or altered mental status.

Common Symptoms:

- Sudden fever, headache, myalgia (especially calf pain)
- Conjunctival suffusion (red eyes), nausea, vomiting
- Jaundice and dark urine in severe disease

Precautions (after suspicion/diagnosis):

- Start antibiotics promptly when leptospirosis suspected (empiric therapy in high-suspicion cases).
- Supportive care, monitor renal function, electrolytes, and liver function.
- Avoid NSAIDs in severe cases with bleeding risk.

Prevention:

- Avoid contact with contaminated water/soil; protective clothing for at-risk workers.
- Rodent control; prophylactic doxycycline in some high-risk exposures per local guidance.

Medications_and_Dosage (general reference, not a prescription):

- Doxycycline (mild cases or prophylaxis):
    - Adults: 100 mg twice daily for 7 days (treatment) or single 200 mg as prophylaxis in some exposure scenarios — follow local guidelines.

    - Children: specialist guidance; doxycycline use in short courses may be considered in older children.
- Severe disease:
    - IV penicillin G (e.g., 1.5–3 million units IV every 6 hours) or IV ceftriaxone/cefotaxime per hospital protocol.
    - Supportive management for renal or hepatic impairment.

#END#

Disease: Measles (Rubeola)

Overview:

- Highly contagious viral illness causing fever, cough, coryza, conjunctivitis, and a characteristic maculopapular rash.
- Complications include otitis media, pneumonia, encephalitis; severe disease more common in malnourished children and immunocompromised individuals.
- Transmission via respiratory droplets; high secondary attack rates.
- Prevention primarily via MMR (measles-containing) vaccination.
- Seek urgent care for high fever, severe respiratory distress, neurologic signs, or dehydration.

Common Symptoms:

- High fever, cough, runny nose, red watery eyes
- Koplik spots (small white spots on buccal mucosa) early
- Diffuse maculopapular rash starting on face and spreading downward

Precautions (after suspicion/diagnosis):

- Airborne/isolation precautions; notify public health for contact tracing.
- Supportive care, vitamin A supplementation in children where indicated (reduces complications).
- Monitor for secondary bacterial infections.

Prevention:

- Two-dose measles-containing vaccine (MMR) per immunization schedules.
- Rapid identification and isolation of cases to prevent outbreaks.

Medications_and_Dosage (general reference, not a prescription):

- Supportive care; no specific antiviral routinely used.
- Vitamin A (reduces morbidity/mortality in children):
    - Children 6–12 months: 100,000 IU orally once daily for 2 days.

    - Children 12 months and older: 200,000 IU orally once daily for 2 days.
    - Adults: provider-directed dosing in severe cases.
- Manage fever and hydration per age-appropriate doses.

#END#

Disease: Mumps

Overview:

- Viral infection (paramyxovirus) that commonly causes parotid gland swelling; can cause orchitis, oophoritis, meningitis, or pancreatitis.
- Transmission by respiratory droplets; vaccination has greatly reduced incidence.
- Most cases are self-limited but complications (especially in post-pubertal males) can include testicular pain and possible infertility (rare).
- Seek care for severe headache, high fever, severe abdominal pain, or testicular swelling.

Common Symptoms:

- Fever, malaise, headache
- Painful swelling of parotid glands (one or both)
- Earache, difficulty chewing or swallowing

Precautions (after suspicion/diagnosis):

- Isolate until swelling subsides per local guidance.
- Symptomatic care, analgesia, and scrotal support if orchitis occurs.
- Notify public health if outbreaks suspected.

Prevention:

- MMR vaccination per routine immunization schedules.

Medications_and_Dosage (general reference, not a prescription):

- Supportive care: analgesics (paracetamol/ibuprofen) per age dosing above.
- No specific antiviral therapy for uncomplicated mumps; manage complications (e.g., hospitalize for meningitis).

#END#

Disease: Chickenpox (Varicella)

Overview:

- Primary infection with varicella-zoster virus causing vesicular, pruritic rash and systemic symptoms.

- Usually mild in children but can be severe in neonates, pregnant people, and immunocompromised patients.
- Transmission via respiratory droplets and contact with lesions.
- Antiviral therapy (acyclovir) indicated for high-risk patients or severe disease; vaccination prevents most cases.
- Seek urgent care for signs of pneumonia, encephalitis, severe dehydration, or secondary bacterial skin infection.

Common Symptoms:

- Fever, malaise, headache
- Pruritic vesicular rash in crops (macules → papules → vesicles → crusts)
- Rash often more severe in adults

Precautions (after suspicion/diagnosis):

- Isolate until lesions crust over to prevent spread.
- Keep skin clean, trim nails to reduce secondary bacterial infection; calamine or antihistamines for pruritus.
- Antiviral therapy for adults, adolescents, pregnant women, neonates, or immunocompromised as indicated.

Prevention:

- Varicella vaccine (two-dose schedule) provides strong protection.
- Varicella-zoster immune globulin for high-risk exposures in some cases.

Medications_and_Dosage (general reference, not a prescription):

- Acyclovir:
    - Adults (treatment): 800 mg orally five times daily for 7 days (typical regimen for otherwise healthy adults when indicated).
    - Children: 20 mg/kg per dose orally four times daily (max dosing per label) for 5–7 days; early initiation within 24–72 hours of rash onset most effective.
    - IV acyclovir for severe disease: dosing per body weight, specialist-guided.
- Symptomatic: paracetamol/antihistamines per ages above; avoid aspirin in children.

#END#

Disease: Pertussis (Whooping Cough)

Overview:

- Highly contagious bacterial respiratory infection caused by Bordetella pertussis.
- Classic severe paroxysmal cough with inspiratory “whoop” in children; young infants may present with apnea.

- Vaccination (DTaP/Tdap) reduces incidence but immunity wanes; early antibiotic therapy reduces transmission.
- Complications include pneumonia, seizures, and apnea in infants.
- Seek urgent care for apnea, cyanosis, severe respiratory distress, or feeding difficulties.

Common Symptoms:

- Prolonged cough with paroxysms, post-tussive vomiting
- Inspiratory whoop (often in children)
- Apnea, especially in infants; low-grade fever may be present

Precautions (after suspicion/diagnosis):

- Macrolide antibiotics for patient and close contacts (post-exposure prophylaxis).
- Isolate until completion of appropriate antibiotic therapy.
- Monitor infants closely; hospitalize when unstable.

Prevention:

- Routine childhood immunization (DTaP) and adult booster (Tdap) per schedules.
- Early treatment and prophylaxis of close contacts.

Medications_and_Dosage (general reference, not a prescription):

- Azithromycin:
    - Adults: 500 mg day 1 then 250 mg once daily days 2–5 (typical regimen).
    - Infants and children: weight-based dosing per label (e.g., 10 mg/kg day 1 then 5 mg/kg days 2–5); follow paediatric guidelines.
    - Older adults: use standard dosing; consider interactions.
- Alternatives: clarithromycin or erythromycin (erythromycin less favoured in infants due to pyloric stenosis concerns).

#END#

Disease: HIV/AIDS (Human Immunodeficiency Virus Infection)

Overview:

- Viral infection that progressively impairs immune function (CD4 T-cells), leading to opportunistic infections and certain malignancies if untreated.
- Transmission via blood, sexual contact, and perinatal exposure.
- Modern antiretroviral therapy (ART) suppresses viral load, restores immunity, and prevents transmission.
- Early diagnosis and lifelong ART improve outcomes; monitoring includes viral load and CD4 counts.

- Seek urgent care for severe opportunistic infections, acute seroconversion illnesses with high fever, or new neurologic deficits.

Common Symptoms:

- Acute infection: fever, sore throat, rash, lymphadenopathy (flu-like).
- Chronic untreated: weight loss, opportunistic infections, persistent fevers, night sweats.
- Many are asymptomatic early on.

Precautions (after suspicion/diagnosis):

- Prompt testing, baseline labs, counsel on transmission prevention.
- Start ART as recommended; adherence counselling and monitoring for drug interactions and side effects.
- Vaccinate with non-live vaccines per guidelines; avoid live vaccines in advanced immunosuppression.

Prevention:

- Condoms, pre-exposure prophylaxis (PrEP) for high-risk uninfected individuals, harm reduction (needle programs), safe blood practices, antenatal screening and peripartum prophylaxis to prevent vertical transmission.

Medications_and_Dosage (general reference, not a prescription):

- Antiretroviral therapy (examples of common regimens — specialist-guided and individualized):
    - Integrase strand transfer inhibitor (INSTI)-based regimens, e.g., Bictegravir/tenofovir alafenamide/emtricitabine:
        - Adults: fixed-dose combination once daily as per product label.
        - Children: paediatric dosing varies by weight/age; specialist-prescribed.
    - Single-tablet regimens exist; choice depends on comorbidities, resistance, and interactions.
- Post-exposure prophylaxis (PEP):
    - Adults: typically, a 28-day course of combination ART started ASAP after exposure; specialist guidance required.
- PrEP:
    - Tenofovir disoproxil fumarate/emtricitabine: one tablet once daily for eligible adults per local guidelines.

#END#

Disease: Asthma (Chronic Airway Inflammatory Disease)

Overview:

- Chronic inflammatory disorder of the airways characterized by variable airflow obstruction and bronchial hyperresponsiveness.
- Symptoms vary over time and may be triggered by allergens, exercise, infections, or irritants.
- Control assessed by symptoms, reliever use, and exacerbation frequency; inhaled corticosteroids are mainstay for long-term control.
- Exacerbations can be life-threatening; action plans and inhaler technique are essential.
- Seek urgent care for severe breathlessness, inability to speak full sentences, cyanosis, or peak flow very low.

Common Symptoms:

- Wheeze, shortness of breath
- Chest tightness, cough (often worse at night/early morning)
- Exercise-induced symptoms

Precautions (after suspicion/diagnosis):

- Ensure a written asthma action plan and correct inhaler technique.
- Use short-acting beta-agonist reliever (SABA) for acute symptoms; escalate therapy per plan.
- Avoid triggers and monitor control; seek early treatment for exacerbations.

Prevention:

- Allergen avoidance where relevant, smoking cessation, influenza vaccination, and good air quality.
- Regular controller therapy (inhaled corticosteroid ± LABA) as indicated.

Medications_and_Dosage (general reference, not a prescription):

- Salbutamol/Albuterol (SABA) inhaler:
    - Adults: 1–2 puffs (100–200 mcg per puff) every 4–6 h PRN; in acute settings 2–6 puffs q20 min for 3 doses may be used.
    - Children: 1–2 puffs via spacer; dosing and spacer use per age.
- Inhaled corticosteroids (e.g., budesonide):
    - Adults: 200–800 mcg per day in divided doses depending on severity.
    - Children: lower doses per label (e.g., 100–400 mcg/day) adjusted by age/weight.
- For exacerbations: oral prednisolone/prednisone 40–50 mg once daily for adults for 3–5 days (short course) per clinician; children dose ~1–2 mg/kg (max ~40 mg) as directed.

#END#

Disease: Chronic Obstructive Pulmonary Disease (COPD)

Overview:

- Progressive airway disease (chronic bronchitis and emphysema) usually due to long-term exposure to noxious particles, especially cigarette smoke.
- Characterized by persistent airflow limitation, dyspnoea, chronic cough, and sputum production.
- Exacerbations accelerate decline; management includes smoking cessation, bronchodilators, inhaled steroids in selected patients, pulmonary rehab, and oxygen therapy when indicated.
- Vaccination and prompt treatment of infections reduce exacerbations.
- Seek urgent care for severe exacerbation with worsening breathlessness, cyanosis, or respiratory failure.

Common Symptoms:

- Progressive dyspnea on exertion
- Chronic productive cough
- Wheeze, frequent respiratory infections

Precautions (after suspicion/diagnosis):

- Smoking cessation is paramount.
- Optimize inhaler therapy, monitor oxygen saturation; acute exacerbations may require antibiotics and steroids.
- Pulmonary rehabilitation and vaccination to reduce exacerbations.

Prevention:

- Avoid smoking and environmental pollutants; occupational protections.
- Influenza and pneumococcal vaccines per guidelines.

Medications_and_Dosage (general reference, not a prescription):

- Short-acting bronchodilator: Salbutamol (SABA) inhaler PRN per doses above.
- Long-acting bronchodilators:
    - Tiotropium (LAMA): Adults one inhalation (18 mcg) once daily via HandiHaler resp. device (product-specific).
    - LABA (e.g., salmeterol) dosing per product label; combination inhalers (LABA/LAMA or LABA/ICS) used per severity.
- Systemic corticosteroid for exacerbation:
    - Prednisone 40 mg once daily for 5 days in adults often used; pediatric doses per specialist if needed.

- Long-term oxygen therapy for chronic hypoxemia as prescribed (target saturations and flow individualized).

#END#

Disease: Acute Bronchitis

Overview:

- Acute inflammation of the bronchi, most commonly viral; usually self-limited (~1–3 weeks).
- Presents with cough (often productive), sometimes with wheeze or low-grade fever.
- Antibiotics are generally not indicated unless bacterial cause suspected or high risk for complications.
- Symptomatic treatment and patient education are mainstays.
- Seek care for high fever, shortness of breath, haemoptysis, or symptoms suggesting pneumonia.

Common Symptoms:

- Persistent cough (may produce sputum)
- Wheeze, chest discomfort
- Low-grade fever, malaise

Precautions (after suspicion/diagnosis):

- Symptomatic treatment: fluids, analgesics, cough suppressants if appropriate for adults.
- Avoid unnecessary antibiotics.
- Monitor for signs of pneumonia or worsening.

Prevention:

- Hand hygiene, respiratory etiquette, vaccination for influenza where appropriate.

Medications_and_Dosage (general reference, not a prescription):

- Supportive care: paracetamol/ibuprofen per ages above.
- Bronchodilator (SABA) trial if wheeze or bronchospasm (salbutamol inhaler 1–2 puffs PRN).
- Antibiotics only if bacterial infection suspected per clinician.

#END#

Disease: Pleural Effusion (symptom/condition, various causes)

Overview:

- Accumulation of fluid in the pleural space from diverse causes (heart failure, infection/empyema, malignancy, pulmonary embolism).
- Presents with dyspnoea, pleuritic chest pain, diminished breath sounds, and reduced chest expansion.
- Diagnosis via chest imaging and thoracentesis to analyse fluid for diagnosis and relieve symptoms.
- Management directed at underlying cause; large effusions may require drainage.
- Seek urgent care for severe dyspnoea, hypoxia, or signs of empyema/sepsis.

Common Symptoms:

- Progressive shortness of breath
- Pleuritic chest pain, cough
- Reduced breath sounds and dullness to percussion on exam

Precautions (after suspicion/diagnosis):

- Diagnostic thoracentesis with fluid analysis (diagnostic and therapeutic).
- If empyema suspected, antibiotics and drainage required.
- Monitor oxygenation and treat underlying cause.

Prevention:

- Manage underlying conditions (heart failure control, infection prevention, cancer care) to reduce risk.

Medications_and_Dosage (general reference, not a prescription):

- Treatment is cause-specific:
    - Heart failure-related effusion: optimize diuretics (e.g., furosemide dosing individualized).
    - Empyema: IV antibiotics guided by cultures (e.g., broad-spectrum until directed) and drainage — dosing per hospital protocols.
- Analgesia and oxygen as needed.
- #END#

Disease: Pulmonary Embolism (PE)

Overview:

- Occlusion of pulmonary arteries by thrombus (often from deep venous thrombosis) leading to ventilation–perfusion mismatch, hypoxia, and possible hemodynamic instability.
- Presents variably from mild dyspnoea to sudden collapse; can be life-threatening.

- Diagnosis by clinical assessment, D-dimer, and imaging (CT pulmonary angiography); treat promptly with anticoagulation or thrombectomy for massive PE.
- Seek immediate emergency care for sudden unexplained shortness of breath, chest pain, syncope, or haemoptysis.

Common Symptoms:

- Sudden onset dyspnoea
- Pleuritic chest pain, cough, sometimes haemoptysis
- Tachycardia, syncope, hypotension in massive PE

Precautions (after suspicion/diagnosis):

- Immediate anticoagulation unless contraindicated; evaluate for bleeding risks.
- Hemodynamically unstable patients may need thrombolysis or embolectomy.
- Investigate and manage DVT source and risk factors; consider IVC filter only in selected cases.

Prevention:

- DVT prophylaxis in high-risk hospitalized patients (mechanical and pharmacologic).
- Early mobilization after surgery, compression stockings, and anticoagulant prophylaxis when indicated.

Medications_and_Dosage (general reference, not a prescription):

- Anticoagulation typical options:
    - Low molecular weight heparin (e.g., enoxaparin):
        - Adults: prophylactic or therapeutic dosing per weight (e.g., therapeutic 1 mg/kg subcutaneously every 12 hours) — hospital dosing and renal function adjustments required.
    - Direct oral anticoagulants (DOACs) for treatment (e.g., apixaban, rivaroxaban):
        - Apixaban adults: 10 mg twice daily for 7 days then 5 mg twice daily (example regimen for PE/DVT) — follow product label.
        - Children: paediatric anticoagulation specialist-guided.
    - Warfarin: loading and maintenance dosing individualized with INR monitoring.
- Thrombolysis (e.g., alteplase) for massive PE per emergency protocol.
- #END#

Disease: Pneumothorax (Spontaneous & Traumatic)

Overview:

- Air in the pleural space causing partial or complete lung collapse; can be spontaneous (primary or secondary) or due to trauma/procedures.
- Presents with sudden pleuritic chest pain and shortness of breath; tension pneumothorax is life-threatening with mediastinal shift and hemodynamic compromise.
- Diagnosis via chest X-ray or chest ultrasound; management ranges from observation to needle decompression/chest tube.
- Seek immediate emergency care for sudden chest pain with severe breathlessness or hypotension.

Common Symptoms:

- Sudden unilateral pleuritic chest pain
- Dyspnoea, reduced breath sounds on affected side
- Hypoxia, hypotension in tension pneumothorax

Precautions (after suspicion/diagnosis):

- Tension pneumothorax requires immediate needle decompression followed by chest tube.
- Small, stable primary spontaneous pneumothorax may be observed with oxygen and follow-up imaging.
- Avoid air travel and high-altitude exposure until resolved per clinician.

Prevention:

- Avoid smoking, counsel patients with underlying lung disease; safety measures to prevent thoracic trauma.

Medications_and_Dosage (general reference, not a prescription):

- Analgesia (paracetamol/NSAIDs per age dosing) for pain.
- Definitive management procedural (needle decompression, chest tube insertion) — no specific medication replaces intervention.
- Oxygen supplementation as needed.

#END#

Disease: Interstitial Lung Disease (ILD) / Pulmonary Fibrosis

Overview:

- A group of chronic lung disorders characterized by inflammation and scarring of lung tissue.
- Leads to stiffness of the lungs, reducing oxygen exchange.
- Causes include autoimmune diseases, occupational exposures, infections, or idiopathic.

- Progressive in nature, often leading to respiratory failure if untreated.
- Prognosis varies by cause and treatment response.

Common Symptoms:

- Shortness of breath (especially with exertion)
- Dry cough
- Fatigue and weakness
- Weight loss
- Clubbing of fingers in advanced cases

Precautions:

- Avoid smoking and environmental exposures (dust, chemicals).
- Regular follow-up with pulmonary specialist.
- Monitor oxygen saturation at home.
- Early management of respiratory infections.

Prevention:

- Use protective equipment in workplaces with dust/chemicals.
- Early treatment of autoimmune and inflammatory conditions.
- Vaccination (influenza, pneumococcal) to prevent infections.

Medications and Dosage:

- Pirfenidone: Adults 801 mg three times daily with food; not recommended in children; elderly dose same as adults with liver monitoring.
- Nintedanib: Adults 150 mg orally twice daily; not recommended in children; adjust dose for tolerability in elderly.
- Oxygen therapy as prescribed.
- Corticosteroids (e.g., Prednisone 0.5–1 mg/kg/day for inflammatory ILD, taper as directed).

#END#

Disease: Bronchiectasis

Overview:

- A chronic condition where airways become abnormally widened due to infection or inflammation.
- Causes impaired clearance of mucus, leading to recurrent infections.
- Can be congenital (e.g., cystic fibrosis) or acquired.
- Progresses slowly but leads to chronic lung damage.

- Early treatment prevents frequent exacerbations.

Common Symptoms:

- Persistent cough with thick, purulent sputum
- Recurrent chest infections
- Wheezing and breathlessness
- Fatigue
- Haemoptysis (in severe cases)

Precautions:

- Perform daily airway clearance techniques (physiotherapy, devices).
- Stay hydrated to thin mucus.
- Prompt treatment of infections with antibiotics.
- Avoid smoking and air pollutants.

Prevention:

- Vaccinations (influenza, pneumococcal).
- Early treatment of childhood lung infections.
- Avoid exposure to lung irritants.

Medications and Dosage:

- Azithromycin (chronic prophylaxis): Adults 250 mg daily or 500 mg 3 times/week; Children 10 mg/kg 3 times/week; elderly monitor cardiac risk.
- Inhaled bronchodilators: e.g., Salbutamol 2 puffs every 4–6 h as needed.
- Inhaled hypertonic saline (3–7%) for mucus clearance.
- Antibiotics: chosen based on sputum culture (dose varies by drug).

#END#

Disease: Lung Cancer (Primary)

Overview:

- Malignant growth of lung tissue, most commonly due to smoking.
- Two main types: Non-small cell (85%) and small cell lung cancer.
- Often diagnosed late due to nonspecific early symptoms.
- Can metastasize rapidly to brain, liver, or bones.
- Prognosis depends on stage and treatment availability.

Common Symptoms:

- Chronic cough

- Haemoptysis (blood in sputum)
- Chest pain
- Shortness of breath
- Unexplained weight loss and fatigue

Precautions:

- Seek medical evaluation for persistent cough or haemoptysis.
- Regular screening for high-risk individuals (e.g., smokers).
- Adhere to treatment plan (surgery, chemo, radiotherapy, immunotherapy).

Prevention:

- Avoid smoking and second-hand smoke.
- Reduce exposure to asbestos, radon, and other carcinogens.
- Maintain a healthy lifestyle with antioxidant-rich diet.

Medications and Dosage:

- Chemotherapy (Cisplatin, Carboplatin, Paclitaxel): doses vary by body surface area and protocol.
- Targeted therapy (Erlotinib 150 mg daily in EGFR-positive cases).
- Immunotherapy (Pembrolizumab 200 mg IV every 3 weeks in selected patients).
- Supportive meds: Morphine for pain (start with 5–10 mg every 4 h in adults, adjust by age and tolerance).

#END#

Disease: Obstructive Sleep Apnoea (OSA)

Overview:

- A sleep disorder characterized by repeated episodes of airway obstruction during sleep.
- Causes intermittent hypoxia and sleep fragmentation.
- Risk factors: obesity, large tonsils, craniofacial abnormalities, alcohol use.
- Associated with hypertension, heart disease, and stroke if untreated.
- Can be mild, moderate, or severe based on apnea-hypopnea index.

Common Symptoms:

- Loud snoring
- Witnessed apneas (breathing stops during sleep)
- Daytime sleepiness

- Morning headaches
- Poor concentration

Precautions:

- Weight reduction if overweight.
- Avoid alcohol and sedatives before sleep.
- Sleep on the side instead of supine.
- Use CPAP device as prescribed.

Prevention:

- Maintain healthy body weight.
- Regular exercise and good sleep hygiene.
- Avoid smoking and alcohol.

Medications and Dosage:

- CPAP (Continuous Positive Airway Pressure): standard treatment across age groups.
- Modafinil for excessive daytime sleepiness: Adults 200 mg once daily in morning; not recommended for children; elderly may require lower doses.
- Oxygen therapy for severe cases not tolerating CPAP.

#END#

Disease: Cystic Fibrosis (Pulmonary Manifestations)

Overview:

- A genetic disorder causing defective chloride channels, leading to thick mucus in lungs.
- Recurrent lung infections and progressive lung damage are common.
- Autosomal recessive inheritance, diagnosed often in childhood.
- Also affects pancreas, GI tract, and reproductive system.
- Life expectancy improved with modern therapies but remains shortened.

Common Symptoms:

- Chronic productive cough
- Recurrent chest infections
- Wheezing and shortness of breath
- Poor weight gain despite good appetite
- Clubbing of fingers

Precautions:

- Daily chest physiotherapy and airway clearance.
- Adequate hydration and high-calorie diet.
- Prompt antibiotic treatment for infections.
- Regular specialist follow-up.

Prevention:

- Genetic counselling for families at risk.
- Vaccinations (influenza, pneumococcal).
- Avoidance of smoking and pollutants.

Medications and Dosage:

- Dornase alfa inhalation: Adults and children >5 y: 2.5 mg once daily.
- Inhaled hypertonic saline (3–7% twice daily).
- Azithromycin prophylaxis: Adults 250–500 mg 3 times/week; Children 10 mg/kg 3 times/week.
- CFTR modulators (Ivacaftor: Adults/children ≥6 y, 150 mg orally every 12 h; dose adjusted in liver disease).
- Pancreatic enzyme replacement: weight-based dosing with meals.

#END#

Disease: Upper Airway Obstruction (Epiglottitis / Laryngeal Causes)

Overview:

- A potentially life-threatening blockage of airflow in the upper airway.
- Common causes: epiglottitis (infection, often Haemophilus influenzae type b), foreign body aspiration, trauma, tumours.
- Rapid swelling can lead to complete airway obstruction.
- Requires immediate medical attention.
- More common in children but can occur at any age.

Common Symptoms:

- Sudden onset of stridor (high-pitched breathing sound)
- Severe sore throat and difficulty swallowing
- Drooling, inability to speak
- Respiratory distress (laboured breathing, cyanosis)
- Fever (in infectious cases)

Precautions:

- Do not attempt to examine the throat in suspected epiglottitis (may trigger obstruction).
- Seek emergency care immediately.
- Maintain upright position for easier breathing.
- Ensure emergency airway management availability.

Prevention:

- Vaccination against Hib (Haemophilus influenzae type b).
- Supervised eating in young children to prevent choking.
- Avoid small objects/toys that may cause aspiration.

Medications and Dosage:

- Empiric antibiotics (Ceftriaxone: Adults 1–2 g IV once daily; Children 50–75 mg/kg/day IV).
- Corticosteroids (Dexamethasone: Adults 4–10 mg IV; Children 0.15 mg/kg IV every 6 h).
- Epinephrine nebulization for severe swelling (dose per weight, emergency use only).
- Oxygen therapy as supportive care.

#END#

Disease: Pulmonary Hypertension (Grouped Causes)

Overview:

- A chronic condition characterized by abnormally high blood pressure in pulmonary arteries.
- Classified into 5 groups: idiopathic, left-heart disease, lung diseases, chronic thromboembolism, and miscellaneous causes.
- Leads to progressive right heart failure if untreated.
- Diagnosis made via echocardiography and right heart catheterization.
- Prognosis depends on early detection and treatment.

Common Symptoms:

- Exertional shortness of breath
- Fatigue, dizziness, syncope
- Chest pain
- Swelling in legs and abdomen (right heart strain)
- Cyanosis in advanced cases

Precautions:

- Avoid high altitudes and excessive exertion.
- Oxygen therapy if hypoxic.
- Regular monitoring with a cardiopulmonary specialist.
- Strict adherence to medications.

Prevention:

- Early treatment of underlying lung/heart diseases.
- Prevent recurrent pulmonary embolism (anticoagulation).
- Avoid smoking and appetite suppressants.

Medications and Dosage:

- Sildenafil (PDE5 inhibitor): Adults 20 mg orally three times daily; not recommended in children unless specialist directed.
- Bosentan (endothelin receptor antagonist): Adults 62.5 mg twice daily → increase to 125 mg twice daily; Children dosing specialist guided.
- Epoprostenol (IV prostacyclin): initiated in hospital, dose titrated individually.
- Diuretics (Furosemide: Adults 20–40 mg daily; Children 1 mg/kg/dose).

#END#

Disease: Respiratory Failure (Acute & Chronic Decompensation)

Overview:

- A condition where lungs fail to maintain adequate oxygen and/or carbon dioxide exchange.
- Acute causes: ARDS, pneumonia, pulmonary embolism, asthma exacerbation.
- Chronic causes: COPD, ILD, neuromuscular disorders.
- Divided into hypoxemic (low O2) and hypercapnic (high CO2).
- Requires urgent hospital management.

Common Symptoms:

- Severe shortness of breath
- Cyanosis (bluish skin/lips)
- Confusion, agitation, or drowsiness
- Rapid or laboured breathing
- Fatigue, inability to speak full sentences

Precautions:

- Seek emergency care immediately.

- Use home oxygen if prescribed.
- Avoid sedatives or narcotics without supervision.
- Vaccinations to prevent infections.

Prevention:

- Control underlying chronic lung and heart diseases.
- Smoking cessation.
- Early treatment of respiratory infections.
- Regular follow-ups for chronic lung patients.

Medications and Dosage:

- Oxygen therapy: titrated to maintain SpO□ >90%.
- Bronchodilators (Salbutamol inhaler 2 puffs every 4–6 h as needed).
- Corticosteroids (Prednisone 40–60 mg/day adults; Children 1–2 mg/kg/day, taper as directed).
- Antibiotics if infection suspected (choice depends on cause).
- Mechanical ventilation in severe cases.

#END#

Disease: Diabetes Mellitus (Type 1 & Type 2)

Overview:

- A chronic metabolic disease marked by high blood sugar levels.
- Type 1: autoimmune destruction of pancreatic beta cells (insulin-dependent).
- Type 2: insulin resistance with relative insulin deficiency.
- Long-term complications: kidney failure, neuropathy, retinopathy, heart disease.
- Diagnosis via fasting glucose, HbA1c, or oral glucose tolerance test.

Common Symptoms:

- Excessive thirst (polydipsia)
- Frequent urination (polyuria)
- Increased appetite (polyphagia)
- Unexplained weight loss (more in Type 1)
- Fatigue and blurred vision

Precautions:

- Monitor blood glucose regularly.
- Maintain a balanced diet and exercise routine.

- Check feet daily for wounds/infections.
- Keep insulin/medications readily available.

Prevention:

- Type 1 cannot be prevented.
- Type 2 prevention: healthy diet, exercise, weight management.
- Regular screening in high-risk individuals.

Medications and Dosage:

- Insulin (Type 1 essential; sometimes Type 2): dosing individualized, typically 0.5–1 unit/kg/day.
- Metformin (Type 2 first-line): Adults start 500 mg once daily → titrate to 1000 mg twice daily; Children ≥10 y: 500 mg once daily → titrate.
- Sulfonylureas (Glimepiride): Adults start 1–2 mg daily → titrate up to 8 mg/day.
- Older adults: cautious dosing due to hypoglycaemia risk.

#END#

Disease: Hypertension (High Blood Pressure)

Overview:

- A chronic condition where blood pressure persistently exceeds normal values.
- Primary (essential) hypertension has no clear cause; secondary may result from kidney or endocrine disease.
- Major risk factor for stroke, heart attack, kidney failure.
- Often asymptomatic, detected during screening.
- Controlled with lifestyle changes and medications.

Common Symptoms:

- Usually asymptomatic
- Sometimes headache, dizziness, blurred vision
- Severe cases: chest pain, shortness of breath, confusion (hypertensive emergency)

Precautions:

- Monitor blood pressure at home.
- Reduce salt intake, exercise regularly.
- Adhere to medication regimen strictly.
- Avoid smoking and excessive alcohol.

Prevention:

- Healthy lifestyle: balanced diet, exercise, weight management.
- Regular screening in adults.
- Stress management.

Medications and Dosage:

- Amlodipine: Adults 5 mg once daily → max 10 mg; Children ≥6 y: 2.5–5 mg daily; elderly start lowers.
- Enalapril: Adults 5–40 mg/day in divided doses; Children 0.08 mg/kg/day; monitor renal function.
- Hydrochlorothiazide: Adults 12.5–25 mg/day; elderly start at low dose; not common in children.

#END#

Disease: Chronic Kidney Disease (CKD)

Overview:

- A progressive loss of kidney function lasting >3 months.
- Common causes: diabetes, hypertension, glomerulonephritis.
- Leads to accumulation of toxins, electrolyte imbalance, anemia, and bone disease.
- Staged by eGFR (G1–G5).
- End-stage kidney disease requires dialysis or transplantation.

Common Symptoms:

- Fatigue and weakness
- Swelling in legs/ankles
- Decreased urine output
- Nausea, vomiting
- High blood pressure
- Itching and muscle cramps

Precautions:

- Regular monitoring of kidney function and blood pressure.
- Limit salt, protein, and potassium intake as advised.
- Avoid nephrotoxic drugs (NSAIDs, contrast dye).
- Fluid intake monitoring.

Prevention:

- Control diabetes and hypertension.

- Healthy diet and regular exercise.
- Avoid smoking and alcohol.
- Early screening for high-risk individuals.

Medications and Dosage:

- ACE inhibitors/ARBs (Enalapril 5–40 mg/day adults; Children 0.1 mg/kg/day) for BP control.
- Erythropoietin (Epoetin alfa 50–100 units/kg weekly SC in adults/children).
- Diuretics (Furosemide 20–80 mg/day adults; Children 1 mg/kg/dose).
- Phosphate binders (Calcium acetate 667 mg with meals adults).

#END#

Disease: Asthma (Chronic Inflammatory Airway Disease)

Overview:

- A chronic inflammatory disorder of the airways.
- Characterized by recurrent wheezing, breathlessness, chest tightness, and coughing.
- Airflow obstruction is variable and reversible.
- Triggered by allergens, infections, exercise, cold air.
- Can be controlled but not cured.

Common Symptoms:

- Wheezing (whistling breath)
- Cough, worse at night/early morning
- Shortness of breath
- Chest tightness
- Fatigue during flare-ups

Precautions:

- Avoid known allergens and triggers.
- Use inhalers regularly as prescribed.
- Monitor peak flow readings.
- Seek urgent care if symptoms worsen.

Prevention:

- Allergen avoidance (dust mites, pets, pollen).
- Smoking cessation and pollutant avoidance.
- Annual flu vaccination.

- Early treatment of respiratory infections.

Medications and Dosage:

- Short-acting beta-agonist (Salbutamol inhaler: Adults 2 puffs every 4–6 h; Children 1 puff if <12 y).
- Inhaled corticosteroids (Budesonide: Adults 200–400 mcg twice daily; Children 100 mcg twice daily).
- Leukotriene antagonists (Montelukast: Adults 10 mg daily; Children 4–5 mg daily).

#END#

Disease: Acne Vulgaris

Overview:

- A common skin condition due to clogged hair follicles from sebum and dead skin.
- Often worsened by hormonal changes (adolescence).
- Severity ranges from mild comedones to severe nodulocystic acne.
- Can cause scarring if untreated.
- Not contagious.

Common Symptoms:

- Blackheads and whiteheads
- Papules, pustules
- Nodules or cysts (severe)
- Oily skin
- Scarring and pigmentation

Precautions:

- Avoid picking or squeezing lesions.
- Use non-comedogenic skin products.
- Maintain gentle skin hygiene.
- Seek dermatologist care if severe.

Prevention:

- Wash face twice daily with mild cleanser.
- Avoid greasy cosmetics.
- Healthy diet with reduced high glycaemic foods.
- Stress management.

Medications and Dosage:

- Benzoyl peroxide gel (2.5–10% applied once/twice daily).
- Topical retinoids (Adapalene 0.1% at night for adults/children >12 y).
- Oral doxycycline: Adults 100 mg/day; not for children <8 y.
- Isotretinoin: Adults 0.5–1 mg/kg/day (specialist supervision).

#END#

Disease: Atopic Dermatitis (Eczema)

Overview:

- A chronic relapsing skin condition with itching and inflammation.
- Often starts in childhood, linked to family history of allergies/asthma.
- Caused by skin barrier defects and immune hyperreactivity.
- Commonly affects flexural areas (elbows, knees).
- Can impair quality of life.

Common Symptoms:

- Intense itching
- Dry, scaly skin
- Red patches
- Lichenification (thickened skin from scratching)
- Secondary infections (weeping, crusts)

Precautions:

- Avoid scratching and irritants (soaps, wool).
- Keep skin moisturized.
- Use antihistamines if itching is severe.
- Monitor for infection signs.

Prevention:

- Regular emollient use.
- Allergen avoidance.
- Control indoor humidity.
- Breastfeeding may lower risk in infants.

Medications and Dosage:

- Topical corticosteroids (Hydrocortisone 1% cream adults/children twice daily short-term).
- Antihistamines (Cetirizine 10 mg daily adults; 5 mg children).

- Severe: Cyclosporine 3–5 mg/kg/day adults (specialist supervision).

#END#

Disease: Contact Dermatitis (Irritant & Allergic)

Overview:

- Skin inflammation caused by direct contact with irritants or allergens.
- Irritant dermatitis due to chemicals, soaps, detergents.
- Allergic dermatitis involves immune-mediated reaction (nickel, cosmetics).
- Often occupational.
- Usually resolves after removing trigger.

Common Symptoms:

- Red, itchy rash
- Burning/stinging sensation
- Dry, cracked skin
- Blisters in severe cases
- Lesions localized to exposure site

Precautions:

- Identify and avoid offending substance.
- Wear protective gloves/clothing if exposed to irritants.
- Wash skin after exposure.
- Use mild cleansers.

Prevention:

- Barrier creams and gloves in workplaces.
- Avoid fragrances/nickel jewellery if allergic.
- Patch testing for recurrent dermatitis.

Medications and Dosage:

- Topical corticosteroids (Betamethasone 0.05% cream adults; children short-term with mild potency).
- Antihistamines (Loratadine 10 mg adults; 5 mg children).
- Severe cases: Oral Prednisone 0.5–1 mg/kg/day taper.

#END#

Disease: Psoriasis

Overview:

- A chronic autoimmune skin disease with rapid skin cell turnover.
- Plaques of thick, scaly, inflamed skin appear on elbows, knees, scalp.
- Linked with arthritis (psoriatic arthritis).
- Relapsing-remitting course.
- Triggered by infections, stress, certain drugs.

Common Symptoms:

- Silvery-white scaly plaques
- Itching, burning
- Nail pitting/thickening
- Joint pain (if arthritis present)

Precautions:

- Avoid scratching plaques.
- Use moisturizing creams regularly.
- Sunlight exposure in moderation may help.
- Manage stress and infections.

Prevention:

- Avoid triggers (alcohol, smoking, certain meds).
- Maintain healthy lifestyle.
- Early treatment of throat infections (strep can worsen psoriasis).

Medications and Dosage:

- Topical corticosteroids (Clobetasol 0.05% twice daily adults).
- Vitamin D analogues (Calcipotriol cream adults).
- Methotrexate: Adults 7.5–25 mg weekly; Children 0.2–0.7 mg/kg weekly.
- Biologics (Etanercept, Adalimumab specialist use).

Disease: Seborrheic Dermatitis

Overview:

- A chronic inflammatory condition affecting sebaceous gland-rich areas (scalp, face, chest).
- Often linked to Malassezia yeast overgrowth.
- Worsens with stress, cold weather, oily skin.
- Not contagious.

- May overlap with dandruff.

Common Symptoms:

- Greasy, yellowish scales
- Red itchy patches on scalp, eyebrows, nasolabial folds
- Dandruff
- Itching and irritation

Precautions:

- Use medicated shampoos regularly.
- Avoid harsh hair products.
- Keep scalp clean and dry.
- Manage stress.

Prevention:

- Good hygiene and scalp care.
- Balanced diet and stress reduction.
- Regular antifungal shampoo use in recurrent cases.

Medications and Dosage:

- Ketoconazole shampoo 2% twice weekly adults/children >12 y.
- Hydrocortisone 1% cream short-term for flare-ups.
- Selenium sulphide shampoo 1–2% weekly.

#END#

Disease: Rosacea

Overview:

- A chronic inflammatory skin disorder affecting central face.
- Triggered by sunlight, alcohol, hot beverages, stress.
- More common in adults >30 y.
- Not contagious.
- Subtypes include erythematotelangiectatic, papulopustular, ocular rosacea.

Common Symptoms:

- Facial flushing and persistent redness
- Visible blood vessels (telangiectasia)
- Papules/pustules on face

- Eye irritation (ocular rosacea)

Precautions:

- Avoid hot, spicy foods and alcohol.
- Use sunscreen daily.
- Gentle skin care routine.
- Avoid topical steroids unless prescribed.

Prevention:

- Identify and avoid personal triggers.
- Stress management.
- Consistent sun protection.

Medications and Dosage:

- Topical metronidazole 0.75% gel twice daily adults.
- Oral doxycycline 40–100 mg/day adults (not for children <8 y).
- Brimonidine gel 0.33% once daily for redness.

#END#

Disease: Urticaria (Hives) & Angioedema

Overview:

- Sudden itchy, raised welts (hives) or deeper swelling (angioedema).
- Triggers: allergies, infections, medications, stress.
- Acute (<6 weeks) or chronic (>6 weeks).
- Angioedema may affect lips, tongue, throat, causing breathing difficulty.
- Usually self-limiting but may recur.

Common Symptoms:

- Itchy red welts (varying size/shape)
- Swelling of lips/eyelids/hands
- Burning or stinging sensation
- Shortness of breath (in severe cases)

Precautions:

- Avoid known allergens (foods, drugs, insect stings).
- Carry emergency epinephrine if severe allergy.
- Wear medical alert ID.

- Seek emergency care if breathing difficulty occurs.

Prevention:

- Allergen avoidance strategies.
- Keep antihistamines handy for recurrent hives.
- Regular follow-up for chronic cases.

Medications and Dosage:

- Antihistamines (Cetirizine 10 mg daily adults; 5 mg children).
- Loratadine 10 mg daily adults; 5 mg children.
- Severe cases: Prednisone 0.5–1 mg/kg/day short-term.
- Emergency: Epinephrine 0.3 mg IM adults; 0.01 mg/kg children.

#END#

Disease: Scabies

Overview:

- A contagious skin infestation by *Sarcoptes scabiei* mite.
- Spread by close skin-to-skin contact.
- Causes intense itching, especially at night.
- Outbreaks common in crowded living conditions.
- Needs simultaneous treatment of close contacts.

Common Symptoms:

- Intense itching at night
- Burrows (thin greyish lines on skin)
- Rash with small red bumps
- Common sites: between fingers, wrists, elbows, waistline
- Secondary infection from scratching

Precautions:

- Avoid close contact until treated.
- Wash clothes, bedding in hot water.
- Cut nails short and clean regularly.
- Treat all household members simultaneously.

Prevention:

- Maintain personal hygiene.

- Avoid sharing clothes/towels.
- Early treatment in affected individuals.

Medications and Dosage:

- Permethrin 5% cream: Apply over entire body (adults and children >2 mo), leave 8–14 h, repeat after 1 week.
- Ivermectin oral: Adults 200 mcg/kg single dose; Children >15 kg same dose.
- Antihistamines for itching (Cetirizine 10 mg adults; 5 mg children).

#END#

Disease: Tinea (Dermatophytosis) — Corporis, Pedis, Cruris, Versicolor

Overview:

- A group of superficial fungal infections affecting skin, nails, and hair.
- Caused by dermatophytes (*Trichophyton*, *Microsporum*).
- Spread by direct contact with infected persons, animals, or surfaces.
- Common in warm, humid climates.
- Not life-threatening but can be recurrent.

Common Symptoms:

- Ring-shaped, itchy red patches (Tinea corporis = body, cruris = groin, pedis = feet/athlete's foot).
- Scaling, cracking skin between toes (pedis).
- Hypopigmented patches in tinea versicolor.
- Itching and burning sensation.

Precautions:

- Keep affected area clean and dry.
- Avoid scratching to prevent spread.
- Do not share clothes, towels, footwear.
- Treat family members if symptomatic.

Prevention:

- Proper hygiene and dry clothing.
- Use antifungal powders in humid conditions.
- Avoid walking barefoot in public showers.

Medications and Dosage:

- Topical antifungals (Clotrimazole 1% cream twice daily adults/children).

- Oral Terbinafine: Adults 250 mg daily × 2–4 weeks; Children 10–20 kg: 62.5 mg daily.
- Ketoconazole shampoo for versicolor (2% weekly).

#END#

Disease: Onychomycosis (Nail Fungus)

Overview:

- Fungal infection of fingernails or toenails.
- Caused mainly by dermatophytes, sometimes yeasts.
- Risk factors: diabetes, trauma, humid environments.
- Slow progressive course, difficult to eradicate.
- Cosmetic and functional impact.

Common Symptoms:

- Thickened, brittle nails
- Yellowish, whitish, or brown discoloration
- Nail detachment (onycholysis)
- Pain or discomfort in severe cases

Precautions:

- Keep nails trimmed and dry.
- Avoid nail salons with poor hygiene.
- Do not share nail clippers.
- Wear breathable shoes and cotton socks.

Prevention:

- Maintain foot hygiene.
- Antifungal powders for recurrent cases.
- Regular monitoring in diabetics.

Medications and Dosage:

- Oral Terbinafine: Adults 250 mg daily × 6–12 weeks (fingernails) or 12–24 weeks (toenails). Not usually for children.
- Itraconazole: Adults 200 mg daily × 12 weeks.
- Topical Ciclopirox nail lacquer applied daily (mild cases).

#END#

Disease: Cutaneous Candidiasis / Intertrigo

Overview:

- A fungal infection of skin folds caused by *Candida albicans*.
- Common in obese, diabetic, or immunosuppressed patients.
- Moist environments promote growth.
- Can involve armpits, groin, breast folds.
- May become recurrent.

Common Symptoms:

- Red, moist rash with satellite pustules
- Itching and burning
- White curd-like discharge (sometimes)
- Foul odor in severe cases

Precautions:

- Keep skin folds dry.
- Wear loose cotton clothing.
- Control blood sugar in diabetics.
- Avoid prolonged wet clothing.

Prevention:

- Good personal hygiene.
- Weight reduction in obese patients.
- Early treatment of minor rashes.

Medications and Dosage:

- Topical Clotrimazole or Miconazole twice daily adults/children.
- Nystatin cream for resistant cases.
- Oral Fluconazole: Adults 150 mg once weekly × 2–4 weeks; Children 6 mg/kg once.

Disease: Impetigo

Overview:

- A highly contagious bacterial skin infection, common in children.
- Caused by *Staphylococcus aureus* or *Streptococcus pyogenes*.
- Spread through direct contact, scratching, or shared items.
- More common in warm, humid conditions.
- Usually mild but can spread quickly.

Common Symptoms:

- Red sores around mouth, nose, arms, legs
- Honey-coloured crusts
- Itching and irritation
- Swelling of lymph nodes (sometimes)

Precautions:

- Avoid scratching lesions.
- Do not share towels, bedding, clothing.
- Wash hands and nails frequently.
- Keep affected children out of school until treated.

Prevention:

- Good hygiene practices.
- Prompt treatment of cuts and insect bites.
- Avoid overcrowded environments when infected.

Medications and Dosage:

- Topical Mupirocin 2% ointment three times daily adults/children.
- Oral Cephalexin: Adults 500 mg every 6 h × 7 days; Children 25–50 mg/kg/day in divided doses.
- Penicillin allergy: Azithromycin 500 mg daily adults; Children 10 mg/kg/day.

#END#

Disease: Cellulitis & Erysipelas

Overview:

- Acute bacterial infections of skin and subcutaneous tissue.
- Cellulitis involves deeper dermis; erysipelas is superficial with sharp borders.
- Commonly caused by *Streptococcus pyogenes* and *Staphylococcus aureus*.
- Risk factors: diabetes, wounds, venous insufficiency.
- Can spread rapidly and cause systemic infection.

Common Symptoms:

- Red, warm, swollen skin (usually legs/face)
- Pain and tenderness
- Fever, chills
- Enlarged lymph nodes

- Distinct raised borders in erysipelas

Precautions:

- Elevate affected limb.
- Rest and hydrate well.
- Monitor for systemic symptoms.
- Seek urgent care if spreading rapidly.

Prevention:

- Treat wounds promptly.
- Good skin hygiene.
- Control diabetes and vascular conditions.
- Prophylactic antibiotics in recurrent cases.

Medications and Dosage:

- Oral Amoxicillin: Adults 500 mg every 8 h × 7–10 days; Children 25–50 mg/kg/day divided doses.
- Cephalexin: Adults 500 mg every 6 h; Children 25–50 mg/kg/day.
- Severe cases: IV Ceftriaxone 1–2 g/day adults; Children 50 mg/kg/day.

#END#

Disease: Folliculitis & Furuncles (Boils)

Overview:

- Folliculitis is a superficial infection and inflammation of the hair follicles, most commonly caused by *Staphylococcus aureus*, but also by fungi or viruses in some cases.
- It often appears in areas prone to friction, sweating, or shaving such as the scalp, beard, thighs, or buttocks.
- Furuncles, or boils, represent a deeper extension of folliculitis, forming painful pus-filled nodules.
- These can occur singly or as carbuncles (clusters of boils).
- People with weakened immune systems, diabetes, or poor hygiene are more susceptible.
- Spread can occur through direct contact or sharing personal items.
- While usually self-limiting, recurrent furuncles may suggest underlying conditions.

Common Symptoms:

- Red, tender, pus-filled bumps.

- Swelling and localized pain.
- Larger, deeper nodules in furuncles.
- Fever and lymph node swelling in severe cases.

Precautions:

- Maintain strict personal hygiene.
- Avoid squeezing or scratching lesions.
- Use clean razors and avoid sharing towels.

Prevention:

- Proper skin care, antibacterial soaps.
- Manage diabetes and obesity.
- Early treatment of small lesions to prevent progression.

Medications and Dosage:

- Topical Mupirocin 2% ointment twice daily.
- Oral Cephalexin: Adults 500 mg every 6 h × 7 day; Children 25–50 mg/kg/day in divided doses.
- Recurrent infections may require nasal decolonization with Mupirocin.

#END#

Disease: Hidradenitis Suppurativa

Overview:

- A chronic, debilitating inflammatory condition involving the apocrine sweat glands, commonly in the axillae, groin, and buttocks.
- The exact cause is unclear but is believed to involve follicular occlusion, immune dysregulation, and secondary bacterial infection.
- It presents with painful nodules, recurrent abscesses, sinus tract formation, and significant scarring over time.
- Risk factors include smoking, obesity, family history, and mechanical friction.
- Often mistaken for simple boils, leading to delays in diagnosis.
- It has a strong psychological impact due to pain, Odor, and cosmetic disfigurement.
- The disease course is chronic, with relapses and remissions.

Common Symptoms:

- Painful, deep nodules in armpits, groin, perineum.
- Abscesses that rupture with foul-smelling discharge.
- Chronic sinus tract formation and scarring.

- Restricted movement due to pain.

Precautions:

- Avoid tight clothing and friction.
- Quit smoking and reduce body weight.
- Maintain cleanliness of affected areas.

Prevention:

- Early dermatology care to prevent progression.
- Lifestyle modification to reduce flares.
- Long-term antibiotics or biologics in resistant disease.

Medications and Dosage:

- Oral Doxycycline 100 mg twice daily × 3 months.
- Clindamycin + Rifampicin: 300 mg twice daily each × 10 weeks.
- Severe: Adalimumab 40 mg subcutaneous weekly (adults only).
- NSAIDs for pain relief.

#END#

Disease: Herpes Simplex Virus (Oral & Genital Herpes)

Overview:

- Caused by HSV-1 (commonly oral) and HSV-2 (commonly genital), both of which establish lifelong latency in nerve ganglia after the initial infection.
- Transmission occurs via direct skin-to-skin contact, kissing, oral sex, or sexual intercourse.
- Primary infections can be more severe with systemic symptoms like fever and malaise, while recurrences are usually milder.
- Stress, illness, menstruation, or sunlight often trigger reactivation.
- Oral herpes typically presents as “cold sores” around the mouth, while genital herpes presents with painful ulcers.
- It is a chronic, recurrent infection with no permanent cure.
- Psychosocial stigma and risk of transmission complicate management.

Common Symptoms:

- Painful grouped vesicles on lips or genitals.
- Tingling or burning before lesions appear.
- Ulcers after rupture of vesicles.
- Fever and lymphadenopathy in primary infection.

Precautions:

- Avoid sexual/intimate contact during outbreaks.
- Use condoms consistently.
- Do not share utensils, razors, or lip balms.

Prevention:

- Safe sexual practices.
- Daily prophylactic antivirals in recurrent/severe cases.
- Sun protection to prevent oral herpes flares.

Medications and Dosage:

- Acyclovir: Adults 400 mg three times daily × 7–10 days; Children 20 mg/kg (max 400 mg) × 5 times daily.
- Valacyclovir: Adults 1000 mg twice daily × 7–10 days.
- Suppressive therapy: Acyclovir 400 mg twice daily (adults).

#END#

Disease: Herpes Zoster (Shingles)

Overview:

- A reactivation of latent varicella-zoster virus in sensory ganglia, usually decades after childhood chickenpox.
- The risk increases with age and immunosuppression.
- It manifests as a painful, unilateral vesicular rash restricted to a single dermatome.
- The initial pain may precede the rash by several days, often leading to misdiagnosis.
- The infection is usually self-limited but can cause serious complications like postherpetic neuralgia, bacterial superinfection, and vision loss (if ophthalmic branch is affected).
- It is not spread person-to-person as shingles, but active lesions can transmit chickenpox to non-immune individuals.
- Vaccination significantly reduces incidence and severity.

Common Symptoms:

- Tingling or burning pain followed by rash.
- Clustered blisters in a dermatomal distribution.
- Fever, headache, fatigue.
- Postherpetic neuralgia in some patients.

Precautions:

- Cover lesions to prevent transmission.
- Avoid contact with pregnant women and unvaccinated children.
- Rest, hydration, and analgesia.

Prevention:

- Shingles vaccine for adults ≥50 years.
- Prompt initiation of antivirals to reduce complications.

Medications and Dosage:

- Acyclovir: Adults 800 mg five times daily × 7 days; Children 20 mg/kg 4× daily.
- Valacyclovir: Adults 1000 mg three times daily × 7 days.
- Pain: NSAIDs, Gabapentin for neuralgia.

#END#

Disease: Molluscum Contagiosum

Overview:

- A benign viral skin infection caused by the Molluscum contagiosum virus (a poxvirus).
- Common in children, sexually active adults, and immunocompromised individuals.
- Transmission occurs via direct skin-to-skin contact, contaminated objects (towels, toys), or swimming pools.
- Lesions are usually small, painless, and self-limiting, resolving within 6–12 months.
- Atopic dermatitis patients are particularly susceptible due to skin barrier defects.
- Although harmless, lesions can be cosmetically bothersome or itchy.
- In immunocompromised individuals, it can become extensive and persistent.

Common Symptoms:

- Small dome-shaped pearly papules with central umbilication.
- Multiple lesions, often grouped.
- Occasionally itchy or inflamed.

Precautions:

- Avoid scratching to prevent spread.
- Do not share towels, razors, or clothing.
- Cover lesions during contact sports/swimming.

Prevention:

- Good hygiene practices.
- Treat underlying eczema to reduce susceptibility.

- Avoid close contact with infected persons.

Medications and Dosage:

- Often no treatment needed (spontaneous resolution).
- Cryotherapy or curettage for persistent lesions.
- Podophyllotoxin 0.5% topical in adults (not for children).
- Imiquimod cream for resistant lesions in adults.

#END#

Disease: Viral Warts (HPV)

Overview:

- Caused by various strains of human papillomavirus (HPV) infecting skin and mucous membranes.
- Transmission occurs via direct contact, sexual activity, or fomites (e.g., showers, gym equipment).
- Warts are benign but often recurrent due to persistence of HPV in basal keratinocytes.
- Types include common warts (hands, fingers), plantar warts (soles), and genital warts.
- Genital warts are linked to high-risk HPV strains that predispose to cervical and anogenital cancers.
- Immunosuppressed individuals are more prone to multiple or resistant warts.
- Treatment is often prolonged and recurrence is common.

Common Symptoms:

- Rough, raised growths on hands, feet, or genital area.
- Plantar warts: painful lesions on weight-bearing areas.
- Genital warts: soft, clustered growths.

Precautions:

- Do not scratch or pick warts.
- Wear slippers in public showers/pools.
- Safe sex practices to prevent genital wart transmission.

Prevention:

- HPV vaccination for adolescents and young adults.
- Good personal hygiene.
- Avoid sharing footwear or personal items.

Medications and Dosage:

- Salicylic acid topical daily application.
- Cryotherapy every 2–3 weeks until resolution.
- Podophyllotoxin for genital warts: Adults apply twice daily × 3 days.
- Imiquimod cream 3× weekly for 16 weeks (adults).

#END#

Disease: Lichen Planus

Overview:

- Lichen planus is a chronic, immune-mediated inflammatory condition affecting the skin, mucous membranes, nails, and hair.
- The exact cause is unknown but involves T-cell–mediated attack on basal keratinocytes.
- It may be associated with hepatitis C infection, certain medications, or autoimmune conditions.
- Skin lesions are usually itchy, polygonal, purple papules.
- Oral lichen planus manifests as painful white lacy patches, erosions, or ulcers.
- Nail involvement may cause thinning, ridging, or loss.
- The disease can persist for months to years, with relapses.

Common Symptoms:

- Itchy, purple, flat-topped papules (Wickham striae visible).
- Oral: painful erosions, white reticular lesions.
- Nail: ridging, thinning, loss.
- Hair loss in scalp involvement.

Precautions:

- Avoid scratching lesions.
- Identify and stop possible causative drugs.
- Maintain good oral hygiene.

Prevention:

- Regular checkups for oral lesions (precancerous risk).
- Manage comorbid hepatitis C if present.

Medications and Dosage:

- Topical corticosteroids (Clobetasol 0.05% cream twice daily).

- Oral Prednisolone: Adults 0.5–1 mg/kg/day tapered.
- Antihistamines for itching.
- Retinoids in resistant disease.

#END#

Disease: Vitiligo

Overview:

- Vitiligo is a chronic autoimmune condition where melanocytes are destroyed, leading to depigmented skin patches.
- The exact cause is multifactorial: genetic predisposition, autoimmunity, oxidative stress, and neural mechanisms.
- It may be associated with other autoimmune diseases like thyroiditis and type 1 diabetes.
- Depigmentation is more visible in darker skin and may cause significant psychological distress.
- It is not contagious and does not affect general health.
- The course is unpredictable: stable, slowly progressive, or rapid spread.
- Cosmetic and social impact is often severe.

Common Symptoms:

- White patches on skin (face, hands, genitals, extremities).
- Premature greying of scalp hair.
- No pain or itching usually.

Precautions:

- Avoid sunburn on depigmented areas.
- Camouflage creams to reduce stigma.
- Psychological counselling if needed.

Prevention:

- No proven prevention, but early treatment may limit spread.
- Manage associated autoimmune disorders.

Medications and Dosage:

- Topical corticosteroids: Adults apply once daily for limited patches.
- Topical Tacrolimus 0.1% ointment (safe for face).
- Phototherapy (NB-UVB thrice weekly).
- Oral Prednisolone 0.3 mg/kg/day in rapidly spreading cases (short course).

#END#

Disease: Alopecia Areata

Overview:

- Alopecia areata is an autoimmune disease targeting hair follicles, leading to patchy, non-scarring hair loss.
- The condition can affect the scalp, beard, eyelashes, or body hair.
- It is associated with genetic susceptibility and autoimmune comorbidities like thyroid disease.
- The course is unpredictable: some patients recover fully; others progress to total scalp (alopecia totalis) or total body hair loss (alopecia universalis).
- Emotional and psychological impact is significant.
- Nail pitting is seen in some patients.
- It is non-contagious and not due to nutritional deficiency.

Common Symptoms:

- Sudden, round patches of hair loss.
- Exclamation mark hairs at margins.
- Nail pitting in some cases.

Precautions:

- Avoid harsh hair treatments and traction.
- Manage stress and comorbid conditions.

Prevention:

- No proven prevention; early therapy may limit progression.
- Regular follow-up to monitor extent.

Medications and Dosage:

- Intralesional Triamcinolone 2.5–10 mg/mL every 4–6 weeks.
- Topical corticosteroids for children.
- Minoxidil 5% solution daily (adults).
- Oral corticosteroids in rapidly progressive disease (short course).

#END#

Disease: Androgenetic Alopecia (Male & Female Pattern Hair Loss)

Overview:

- A common genetic form of progressive, non-scarring hair loss influenced by androgens.
- In men, it presents as bitemporal recession and vertex baldness.
- In women, diffuse thinning occurs, particularly in the crown, with preserved hairline.
- Caused by increased sensitivity of follicles to dihydrotestosterone (DHT).
- Progresses slowly over years and is permanent without treatment.
- Psychological and social effects are significant.
- Early intervention improves response to therapy.

Common Symptoms:

- Gradual thinning of scalp hair.
- Male: receding hairline, crown baldness.
- Female: central thinning, widening of hair parting.

Precautions:

- Avoid harsh hair treatments and traction.
- Stress management.

Prevention:

- No absolute prevention but early treatment slows progression.

Medications and Dosage:

- Minoxidil topical: Adults 2–5% solution applied twice daily.
- Finasteride: Adult men 1 mg orally daily.
- Spironolactone (for women): 50–100 mg daily.

#END#

Disease: Keloids & Hypertrophic Scars

Overview:

- Abnormal wound healing characterized by excessive fibroblast proliferation and collagen deposition.
- Hypertrophic scars remain within wound boundaries; keloids extend beyond.
- Genetic predisposition (more common in darker skin types).
- Triggered by trauma, surgery, burns, or acne.
- May cause itching, pain, and cosmetic disfigurement.
- Keloids may recur after surgical removal.
- Difficult to treat and often require combination therapy.

Common Symptoms:

- Raised, firm, shiny scars.
- Itching or tenderness.
- Extension beyond wound margins (keloids).

Precautions:

- Avoid unnecessary skin trauma or piercing.
- Monitor healing wounds carefully.

Prevention:

- Pressure dressings post-surgery in predisposed individuals.
- Early treatment of hypertrophic scars to prevent keloids.

Medications and Dosage:

- Intralesional Triamcinolone 10–40 mg/mL every 4–6 weeks.
- Silicone gel sheets applied daily.
- Cryotherapy for small lesions.

#END#

Disease: Pyoderma Gangrenosum

Overview:

- A rare, non-infectious neutrophilic dermatosis presenting as painful ulcers with undermined borders.
- Associated with systemic diseases like inflammatory bowel disease, rheumatoid arthritis, or hematologic malignancies.
- Often misdiagnosed as an infection, leading to inappropriate antibiotic use.
- Pathogenesis involves immune dysfunction with neutrophil overactivity.
- Ulcers expand rapidly and are extremely painful.
- Trauma may worsen lesions (pathergy phenomenon).
- Requires immunosuppressive therapy, not surgery.

Common Symptoms:

- Painful, necrotic ulcers with violaceous borders.
- Rapid progression.
- Associated systemic disease symptoms.

Precautions:

- Avoid trauma to skin.

- Early recognition to prevent ulcer enlargement.

Prevention:

- Control underlying systemic disease.
- Minimize skin injury in predisposed patients.

Medications and Dosage:

- Oral Prednisolone: Adults 0.5–1 mg/kg/day.
- Cyclosporine 3–5 mg/kg/day (adults).
- Biologics (Infliximab, Adalimumab) in refractory cases.

#END#

Disease: Dermatitis Herpetiformis

Overview:

- A chronic, autoimmune blistering skin disorder associated with gluten sensitivity and celiac disease.
- Caused by IgA deposition in dermal papillae.
- Lesions are extremely itchy and symmetrically distributed.
- Common sites: elbows, knees, buttocks, scalp.
- Strong association with gluten-sensitive enteropathy, even in absence of GI symptoms.
- Lifelong gluten-free diet is the cornerstone of management.
- Untreated disease increases risk of intestinal lymphoma.

Common Symptoms:

- Clusters of vesicles with intense itching.
- Symmetrical distribution on extensor surfaces.
- GI symptoms may be absent.

Precautions:

- Avoid scratching to reduce infection risk.
- Gluten-free diet adherence.

Prevention:

- Lifelong strict gluten-free diet.
- Early treatment reduces lymphoma risk.

Medications and Dosage:

- Dapsone: Adults 50–100 mg daily; Children 0.5–1.5 mg/kg/day.

- Gluten-free diet lifelong.
- Sulphapyridine if Dapsone not tolerated.

#END#

Disease: Pityriasis Rosea

Overview:

- A self-limiting papulosquamous skin eruption, likely viral in origin (HHV-6/7 implicated).
- Begins with a single “herald patch” followed by multiple smaller oval lesions.
- Distribution is often along skin cleavage lines (“Christmas tree pattern”).
- Common in adolescents and young adults.
- Usually resolves spontaneously in 6–8 weeks.
- Not contagious and usually mild.
- May be confused with fungal infections or drug eruptions.

Common Symptoms:

- Initial large solitary herald patch.
- Subsequent smaller pink, scaly lesions.
- Mild itching in some patients.

Precautions:

- Avoid harsh soaps and excessive heat.
- Use soothing lotions to reduce itching.

Prevention:

- No specific prevention (self-limiting condition).

Medications and Dosage:

- Antihistamines for itching.
- Topical corticosteroids (low potency).
- Acyclovir 400 mg 5× daily in severe cases (adults).

#END#

Disease: Pityriasis Versicolor (Tinea Versicolor)

Overview:

- A superficial fungal infection caused by *Malassezia* species.
- Presents as hypo- or hyperpigmented scaly patches.

- Common in adolescents and young adults due to sebaceous activity.
- Hot, humid climates promote growth.
- Cosmetic concern is main issue; condition is harmless.
- Often recurs during summer.
- Diagnosis is confirmed by KOH mount showing “spaghetti and meatballs” appearance.

Common Symptoms:

- Irregular hypo/hyperpigmented patches on chest, back, arms.
- Fine scaling.
- Mild itching occasionally.

Precautions:

- Avoid excessive sweating and oily creams.
- Loose cotton clothing.

Prevention:

- Regular antifungal shampoo use in prone individuals.
- Reduce humidity and sweat exposure.

Medications and Dosage:

- Topical Ketoconazole 2% shampoo applied twice weekly × 2–4 weeks.
- Oral Fluconazole: Adults 300 mg weekly × 2–4 weeks; Children 6 mg/kg once weekly.

#END#

Disease: Granuloma Annulare

Overview:

- A benign, chronic skin condition of unknown cause.
- Believed to involve immune-mediated granulomatous inflammation.
- Presents as ring-shaped papules or plaques, commonly on hands, feet, or joints.
- Usually asymptomatic and self-limiting.
- More common in children and young adults.
- Occasionally associated with diabetes, thyroid disease, or dyslipidemia.
- Often resolves spontaneously in months to years.

Common Symptoms:

- Annular, raised, skin-coloured or reddish papules.

- Localized or generalized distribution.
- Usually not itchy or painful.

Precautions:

- Reassurance since benign.
- Avoid unnecessary trauma.

Prevention:

- No specific prevention known.

Medications and Dosage:

- Topical corticosteroids.
- Intralesional Triamcinolone 2.5–5 mg/mL for localized lesions.
- Hydroxychloroquine 200 mg daily (adults) for generalized disease.

#END#

Disease: Seborrheic Keratosis

Overview:

- A common benign epidermal tumour of older adults.
- Caused by proliferation of keratinocytes; not related to sun exposure.
- Lesions are often multiple, brown, “stuck-on” plaques.
- May mimic melanoma or pigmented basal cell carcinoma.
- Often asymptomatic, sometimes itchy.
- No malignant potential, but sudden eruption of multiple lesions (Leser-Trélat sign) may indicate internal malignancy.
- Cosmetic removal is common reason for treatment.

Common Symptoms:

- Waxy, stuck-on papules or plaques.
- Brown, black, or tan lesions.
- Itching occasionally.

Precautions:

- Reassurance of benign nature.
- Monitor for suspicious changes.

Prevention:

- No prevention; age-related.

Medications and Dosage:

- Cryotherapy, curettage, or laser removal for cosmetic reasons.
- No systemic medications required.

#END#

Disease: Actinic Keratosis

Overview:

- A premalignant lesion caused by chronic sun exposure and UV damage.
- Appears as rough, scaly patches on sun-exposed areas (face, scalp, forearms).
- More common in elderly and fair-skinned individuals.
- Has potential to progress to squamous cell carcinoma if untreated.
- A marker of cumulative UV exposure and skin cancer risk.
- Early detection and treatment are important.
- Recurrence is common if sun protection is not followed.

Common Symptoms:

- Rough, sandpaper-like patches.
- Pink or brown scaly lesions.
- May be tender or itchy.

Precautions:

- Regular skin self-examination.
- Strict sun protection (hats, sunscreen).

Prevention:

- Consistent sunscreen use (SPF ≥30).
- Avoid tanning beds and peak sun hours.

Medications and Dosage:

- Cryotherapy for isolated lesions.
- Topical 5-Fluorouracil 5% cream applied twice daily × 2–4 weeks.
- Imiquimod 5% cream 3× weekly × 4 weeks.

#END#

Disease: Basal Cell Carcinoma (BCC)

Overview:

- The most common skin cancer, arising from basal keratinocytes.

- Slow-growing, locally invasive, rarely metastasizes.
- Caused by chronic UV exposure, fair skin, and genetic predisposition.
- Presents as pearly nodules, ulcerated lesions, or pigmented plaques.
- Common sites: face, nose, ears.
- May cause significant local tissue destruction if untreated.
- Excellent prognosis with early treatment.

Common Symptoms:

- Pearly, translucent nodule with telangiectasia.
- Non-healing ulcer (“rodent ulcer”).
- Bleeding or crusting lesions.

Precautions:

- Regular skin checks for suspicious growths.
- Avoid excessive sun exposure.

Prevention:

- Sun protection lifelong.
- Early detection and excision.

Medications and Dosage:

- Surgical excision with margins.
- Mohs micrographic surgery for high-risk sites.
- Imiquimod 5% cream for superficial BCC.

#END#

Disease: Squamous Cell Carcinoma (SCC)

Overview:

- A malignant tumour of keratinocytes, second most common skin cancer.
- Associated with chronic UV exposure, immunosuppression, chronic ulcers, or HPV infection.
- More aggressive than BCC, with potential for lymph node spread.
- Lesions often arise from actinic keratosis.
- Appears as firm, scaly nodules or non-healing ulcers.
- Early detection significantly improves outcome.
- Risk is higher in transplant recipients and immunocompromised patients.

Common Symptoms:

- Scaly, red plaques or nodules.
- Non-healing, ulcerated, painful lesions.
- Enlarged lymph nodes if metastasized.

Precautions:

- Skin monitoring and biopsy of suspicious lesions.
- Avoid chronic irritation or trauma.

Prevention:

- Sun protection.
- HPV vaccination (prevents mucosal SCC).

Medications and Dosage:

- Surgical excision with clear margins.
- Radiation therapy in inoperable cases.
- Systemic chemotherapy in advanced disease.

#END#

Disease: Melanoma

Overview:

- The most dangerous form of skin cancer, arising from melanocytes.
- Strongly associated with intense, intermittent UV exposure and genetic predisposition.
- Lesions show asymmetry, irregular borders, varied colours, diameter >6 mm, and evolution (ABCDE rule).
- High risk of metastasis to lymph nodes, brain, lungs, and liver.
- Early diagnosis and excision are life-saving.
- Family history and atypical mole syndrome increase risk.
- Prognosis depends on depth (Breslow thickness).

Common Symptoms:

- Changing mole with irregular shape or colour.
- New pigmented lesion with asymmetry.
- Itching, bleeding, or ulceration.

Precautions:

- Regular dermatologic checkups.
- Self-monitoring for ABCDE changes.

Prevention:

- Sun protection and avoidance of tanning beds.
- Prophylactic removal of suspicious moles.

Medications and Dosage:

- Wide surgical excision with margins.
- Immunotherapy (Nivolumab, Pembrolizumab).
- Targeted therapy (BRAF/MEK inhibitors in mutation-positive patients).

#END#

Disease: Drug Eruptions

Overview:

- Adverse cutaneous reactions caused by medications.
- Range from mild exanthematous rashes to severe life-threatening conditions like SJS/TEN.
- Common offending drugs: antibiotics (penicillin, sulfa), NSAIDs, anticonvulsants.
- Immune-mediated hypersensitivity is often involved.
- Reactions may appear within days to weeks of starting drug.
- Early recognition and withdrawal of drug are essential.
- Severe cases require hospitalization.

Common Symptoms:

- Widespread red rash.
- Itching, fever.
- Blisters, mucosal erosions in severe cases.

Precautions:

- Always inform doctor about drug allergies.
- Avoid self-medication.

Prevention:

- Careful drug history before prescribing.
- Genetic screening in high-risk drugs (e.g., HLA-B*1502 for carbamazepine).

Medications and Dosage:

- Mild: Antihistamines, topical steroids.
- Severe: Systemic corticosteroids (Prednisolone 0.5–1 mg/kg/day).
- Supportive care in hospital for SJS/TEN.

#END#

Disease: Stevens-Johnson Syndrome (SJS) / Toxic Epidermal Necrolysis (TEN)

Overview:

- Severe, life-threatening mucocutaneous reactions, usually drug-induced.
- Characterized by widespread epidermal necrosis and detachment.
- SJS involves <10% body surface; TEN >30%.
- Drugs commonly responsible: sulphonamides, anticonvulsants, allopurinol, NSAIDs.
- Patients present with flu-like prodrome followed by painful rash, blisters, and mucosal involvement.
- Mortality is high, especially in TEN, due to infection and multi-organ failure.
- Requires ICU or burn unit management.

Common Symptoms:

- Painful erythematous rash.
- Widespread blisters and skin peeling.
- Oral, ocular, genital mucosal erosions.
- Fever, malaise.

Precautions:

- Immediate drug withdrawal.
- Hospitalization in intensive care.

Prevention:

- Genetic screening for high-risk drugs.
- Avoid re-exposure to causative medication.

Medications and Dosage:

- Supportive care (fluids, wound care, infection prevention).
- Systemic corticosteroids: Prednisolone 1–2 mg/kg/day (short course).
- IV Immunoglobulin: 1 g/kg/day × 2–3 days.

#END#

Disease: Cutaneous Lupus Erythematosus

Overview:

- Autoimmune condition where the immune system attacks skin, often associated with systemic lupus erythematosus (SLE).

- Subtypes: discoid lupus erythematosus (DLE), subacute cutaneous lupus, and acute lupus rash.
- Triggers include sunlight, drugs, and infections.
- Lesions may scar and cause pigmentary changes.
- Chronic DLE may lead to scarring alopecia.
- High risk of systemic involvement in some patients.
- Sun sensitivity is a hallmark feature.

Common Symptoms:

- Red, scaly plaques on face, scalp, ears.
- Butterfly rash over cheeks/nose.
- Photosensitivity.

Precautions:

- Avoid sun exposure.
- Regular follow-up for systemic lupus features.

Prevention:

- Strict photoprotection.
- Early treatment to prevent scarring.

Medications and Dosage:

- Topical corticosteroids.
- Hydroxychloroquine 200–400 mg/day (adults).
- Sunscreen SPF ≥50 daily.

#END#

Disease: Eczema Herpeticum

Overview:

- A severe viral skin infection caused by herpes simplex virus in patients with atopic dermatitis.
- Represents widespread, disseminated HSV infection due to impaired skin barrier.
- Considered a dermatologic emergency.
- Lesions spread rapidly and can become secondarily infected with bacteria.
- Systemic involvement (fever, lymphadenopathy) may occur.
- Risk of complications includes keratitis, encephalitis.
- Requires urgent antiviral therapy.

Common Symptoms:

- Clusters of painful, punched-out erosions.
- Fever, malaise.
- Rapid spread of lesions.

Precautions:

- Avoid scratching lesions.
- Seek immediate medical attention.

Prevention:

- Early recognition and treatment of eczema.
- Avoid contact with people with cold sores.

Medications and Dosage:

- IV Acyclovir: Adults 5–10 mg/kg every 8 h; Children 10–15 mg/kg every 8 h.
- Oral Acyclovir for mild cases: 400 mg 3× daily (adults).

#END#

Disease: Necrotizing Fasciitis

Overview:

- A rapidly progressive, life-threatening bacterial soft tissue infection.
- Caused by Group A Streptococcus, anaerobes, or polymicrobial infections.
- Involves fascia and subcutaneous tissue, sparing skin initially.
- Presents with severe pain out of proportion to findings.
- High mortality if untreated.
- Risk factors: diabetes, immunosuppression, trauma, surgery.
- Requires urgent surgery and IV antibiotics.

Common Symptoms:

- Severe pain, swelling, erythema.
- Rapid progression to necrosis, blisters, gangrene.
- Fever, shock.

Precautions:

- Early hospital care for rapidly worsening soft tissue infections.

Prevention:

- Good wound hygiene.

- Prompt treatment of infections.

Medications and Dosage:

- Broad-spectrum IV antibiotics: Piperacillin-Tazobactam 4.5 g IV every 6 h.
- Clindamycin 900 mg IV every 8 h.
- Immediate surgical debridement.

#END#

Disease: Cutaneous Manifestations of Systemic Disease (Dermatomyositis, Xanthomas, etc.)

Overview:

- Many systemic diseases manifest first in the skin, offering early diagnostic clues.
- Dermatomyositis: inflammatory myopathy with characteristic heliotrope rash and Gottron papules. Associated with malignancy.
- Xanthomas: yellow cholesterol-rich deposits, often linked to dyslipidaemia and metabolic syndrome.
- Skin signs may precede systemic symptoms, aiding early recognition.
- Other systemic diseases with skin clues: diabetes (necrobiosis lipoidica), thyroid disease (pretibial myxoedema), liver disease (spider angiomas).
- Identifying these signs helps diagnose hidden systemic illness.
- Requires multidisciplinary care.

Common Symptoms:

- Dermatomyositis: heliotrope rash, Gottron papules.
- Xanthomas: yellowish nodules on eyelids, tendons.
- General systemic disease clues: rashes, pigment changes.

Precautions:

- Regular monitoring for systemic illness.
- Early referral to specialists.

Prevention:

- Control underlying systemic diseases (diabetes, dyslipidaemia).

Medications and Dosage:

- Dermatomyositis: Prednisolone 0.5–1 mg/kg/day.
- Xanthomas: Statins (Atorvastatin 10–40 mg/day adults).
- Supportive topical therapy for skin lesions.

#END#

Nutritional Deficiencies: Causes, Symptoms, and Remedies

Nutritional deficiencies occur when the body does not receive an adequate amount of essential nutrients needed for growth, energy, and overall health. These nutrients include carbohydrates, proteins, fats, vitamins, minerals, and water. Deficiencies can lead to severe health complications, affecting bodily functions and metabolism.

#END#

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1. Carbohydrates

Importance:
Carbohydrates are the primary energy source for the human body. They fuel muscles, the brain, and the nervous system. They are crucial for physical activity and cognitive functions.

Deficiency Effects:
Lack of carbohydrates forces the body to use fat and protein for energy, leading to conditions like **ketosis**, **muscle wasting**, and **mental fatigue**.

Symptoms:

- Weakness and dizziness
- Confusion and poor brain function
- Unintentional weight loss

Sources & Treatment:

- Natural Sources: Rice, wheat, potatoes, corn, bananas, honey
- Supplements: Glucose powders, oral rehydration salts

#END#

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2. Proteins

Importance:
Proteins are the building blocks of life. They form muscles, enzymes, hormones, and play a key role in immunity.

Deficiency Diseases:

- **Kwashiorkor** (protein deficiency with adequate calories)
- **Marasmus** (protein and calorie deficiency)

Symptoms:

- Edema (swelling)
- Stunted growth in children
- Fatigue, poor immunity

Sources & Treatment:

- Natural Sources: Eggs, milk, meat, fish, beans, lentils, soy
- Supplements: Protein powders, amino acid supplements

#END#

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3. Fats

Importance:
Fats provide energy, help absorb fat-soluble vitamins (A, D, E, K), and protect organs.

Deficiency Symptoms:

- Dry, scaly skin
- Brittle hair
- Hormonal imbalance

Sources & Treatment:

- Natural Sources: Olive oil, nuts, seeds, fish, avocados
- Supplements: Omega-3, Omega-6 capsules, cod liver oil

#END#

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4. Water

Importance:
Water is vital for hydration, regulating body temperature, and waste removal.

Deficiency Effects:

- **Dehydration**, kidney stones, constipation

Symptoms:

- Dry mouth
- Low urine output
- Dizziness

Sources & Treatment:

- Natural Sources: Water, watermelon, cucumber, soups
- Medical: Oral rehydration salts (ORS), IV fluids

#END#

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5. Vitamins

Vitamins are organic compounds required in small quantities but essential for metabolic processes and overall health.

#END#

Vitamin A

Importance & Functions:

- Maintains **vision** (especially night vision)
- Supports **immune function**
- Essential for **cell growth and reproduction**
- Keeps **skin and mucous membranes healthy**

Deficiency Diseases:

- **Night blindness**
- **Xerophthalmia** (dryness of eyes)

Symptoms:

- Poor vision in low light
- Dry eyes and rough skin
- Increased risk of infections

Daily Requirement (RDA):

- Men: 900 µg/day
- Women: 700 µg/day

Risk Factors:

- Poor diet lacking vegetables
- Fat malabsorption disorders
- Chronic alcoholism

Natural Sources:
Carrots, spinach, sweet potatoes, liver, eggs, dairy products

Supplements:
Retinol or Vitamin A capsules

#END#

Vitamin B-Complex: Detailed Explanation

The Vitamin B-Complex consists of 8 water-soluble vitamins, each playing a crucial role in energy metabolism, nervous system health, and cell functions. They cannot be

stored in large amounts in the body, so they must be consumed regularly through diet.

#END#

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1. Vitamin B1 (Thiamine)

Importance & Functions:

- **Helps in carbohydrate metabolism to release energy**
- **Supports nerve function and muscle contraction**
- **Maintains heart health**

Deficiency Diseases:

- **Beriberi (dry or wet): Nervous and cardiovascular problems**
- **Wernicke-Korsakoff syndrome (common in alcoholics)**

Symptoms:

- **Fatigue, irritability**
- **Tingling in hands and feet**
- **Muscle weakness, poor reflexes**

Daily Requirement (RDA):

- **Adults: 1.1–1.2 mg/day**

Risk Factors:

- **Chronic alcoholism**
- **High carbohydrate diet without diversity**
- **Long-term diuretic use**

Natural Sources:
Whole grains, nuts, pork, legumes, fortified cereals

Supplements:
Thiamine tablets or injections

#END#

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2. Vitamin B2 (Riboflavin)

Importance & Functions:

- **Helps in energy production and fat metabolism**
- **Essential for healthy skin and eyes**
- **Plays a role in antioxidant functions**

Deficiency Disease:

- **Ariboflavinosis**

Symptoms:

- **Cracks and sores at the corners of the mouth (cheilitis)**
- **Swollen, sore tongue**
- **Skin disorders, eye fatigue**

Daily Requirement:

- **Adults: 1.1–1.3 mg/day**

Risk Factors:

- **Poor diet**
- **Alcoholism**
- **Thyroid disorders**

Natural Sources:
Milk, eggs, green leafy vegetables, almonds, liver

Supplements:
Riboflavin tablets

#END#

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3. Vitamin B3 (Niacin)

Importance & Functions:

- **Supports metabolism of carbohydrates, fats, and proteins**
- **Helps in DNA repair and synthesis**
- **Reduces cholesterol levels**

Deficiency Disease:

- **Pellagra (characterized by the 3 Ds: Dermatitis, Diarrhoea, Dementia)**

Symptoms:

- **Skin rashes in sun-exposed areas**
- **Digestive problems, diarrhoea**
- **Mental confusion, depression**

Daily Requirement:

- **Adults: 14–16 mg/day**

Risk Factors:

- **Corn-based diet (low in niacin)**

- **Alcoholism**
- **Certain digestive disorders**

Natural Sources:
Chicken, tuna, peanuts, mushrooms, whole grains

Supplements:
Niacin tablets (used to lower cholesterol)

#END#

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4. Vitamin B5 (Pantothenic Acid)

Importance & Functions:

- **Crucial for the synthesis of coenzyme A**
- **Supports fatty acid metabolism and energy production**

Deficiency:
Rare, but can cause general weakness

Symptoms:

- **Tingling in hands and feet**
- **Fatigue and irritability**
- **Depression and sleep problems**

Daily Requirement:

- **Adults: 5 mg/day**

Risk Factors:

- **Severe malnutrition**
- **Alcohol dependence**

Natural Sources:
Egg yolk, broccoli, avocados, whole grains, chicken

Supplements:
Pantothenic acid supplements

#END#

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5. Vitamin B6 (Pyridoxine)

Importance & Functions:

- **Helps in amino acid metabolism and neurotransmitter synthesis**
- **Supports haemoglobin production**
- **Important for brain function**

Deficiency Diseases:

- **Microcytic anaemia**
- **Neurological disorders**

Symptoms:

- **Irritability, confusion**
- **Mouth sores**
- **Depression and nerve problems**

Daily Requirement:

- **Adults: 1.3–1.7 mg/day**

Risk Factors:

- **Alcoholism**
- **Kidney disease**
- **Certain medications (e.g., isoniazid)**

Natural Sources:
Bananas, potatoes, chicken, fish, nuts

Supplements:
Pyridoxine tablets

#END#

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6. Vitamin B7 (Biotin)

Importance & Functions:

- **Helps in fatty acid synthesis, glucose production**
- **Maintains healthy hair, skin, and nails**

Deficiency:
Rare, but can occur due to raw egg white consumption (contains avidin)

Symptoms:

- **Hair thinning and loss**
- **Brittle nails**
- **Skin rash around eyes and nose**

Daily Requirement:

- **Adults: 30 µg/day**

Risk Factors:

- **Long-term raw egg consumption**

- **Genetic disorders**

Natural Sources:
Egg yolk, nuts, seeds, salmon, avocado

Supplements:
Biotin capsules

#END#

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7. Vitamin B9 (Folic Acid/Folate)

Importance & Functions:

- **Crucial for DNA synthesis and cell division**
- **Essential during pregnancy to prevent neural tube defects**
- **Helps in red blood cell formation**

Deficiency Diseases:

- **Megaloblastic anaemia**
- **Birth defects (spina bifida)**

Symptoms:

- **Fatigue, irritability**
- **Poor growth**
- **Shortness of breath**

Daily Requirement:

- **Adults: 400 µg/day**
- **Pregnancy: 600 µg/day**

Risk Factors:

- **Poor diet**
- **Alcohol use**
- **Pregnancy without supplementation**

Natural Sources:
Spinach, lentils, asparagus, oranges

Supplements:
Folic acid tablets

#END#

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8. Vitamin B12 (Cobalamin)

Importance & Functions:

- **Maintains nerve cells and aids in red blood cell formation**
- **Involved in DNA synthesis**

Deficiency Diseases:

- **Pernicious anaemia**
- **Neurological disorders**

Symptoms:

- **Weakness, fatigue**
- **Memory loss**
- **Tingling in hands and feet**

Daily Requirement:

- **Adults: 2.4 µg/day**

Risk Factors:

- **Strict vegetarian/vegan diets**
- **Pernicious anaemia (autoimmune condition)**
- **Gastric surgery**

Natural Sources:
Meat, eggs, dairy, fortified cereals

Supplements:
Cyanocobalamin injections or tablets

#END#

Vitamin C (Ascorbic Acid)

Importance & Functions:

- Strong **antioxidant**
- Helps in **collagen synthesis** (important for skin, gums, bones)
- Boosts **immune system**
- Aids in **iron absorption**

Deficiency Disease:

- **Scurvy**

Symptoms:

- Bleeding gums and loose teeth
- Poor wound healing
- Weakness and fatigue

Daily Requirement:

- Men: 90 mg/day
- Women: 75 mg/day

Risk Factors:

- Smoking (increases vitamin C needs)
- Poor fruit and vegetable intake

Natural Sources:
Oranges, guava, strawberries, bell peppers, kiwi

Supplements:
Ascorbic acid tablets or chewable tablets

#END#

Vitamin D

Importance & Functions:

- Regulates **calcium and phosphorus absorption**
- Maintains **bone and teeth health**
- Supports **immune system**

Deficiency Diseases:

- **Rickets** (in children – soft bones)
- **Osteomalacia** (in adults)

Symptoms:

- Bone pain and deformities
- Muscle weakness
- Frequent fractures

Daily Requirement:

- Adults: 600–800 IU/day

Risk Factors:

- Lack of sunlight exposure
- Vegan diet without fortified foods
- Chronic kidney or liver disease

Natural Sources:
Sunlight exposure, fortified milk, eggs, fish (salmon, sardines)

Supplements:
Vitamin D3 (cholecalciferol) tablets or drops

#END#

Vitamin E

Importance & Functions:

- Acts as an **antioxidant**, protecting cells from damage
- Supports **immune function**
- Improves **skin health**

Deficiency Disease:

- Rare, but can cause **neurological and muscle damage**

Symptoms:

- Weak muscles
- Vision problems
- Poor coordination

Daily Requirement:

- Adults: 15 mg/day

Risk Factors:

- Fat absorption disorders (cystic fibrosis, Crohn's disease)
- Very low-fat diets

Natural Sources:
Sunflower oil, almonds, peanuts, spinach, avocados

Supplements:
Tocopherol capsules

#END#

Vitamin K

Importance & Functions:

- Essential for **blood clotting**
- Maintains **bone health**

Deficiency Diseases:

- **Bleeding disorders**

Symptoms:

- Easy bruising
- Prolonged bleeding from wounds

- Nosebleeds

Daily Requirement:

- Men: 120 µg/day
- Women: 90 µg/day

Risk Factors:

- Long-term antibiotic use (destroys gut bacteria)
- Liver disease

Natural Sources:
Kale, spinach, broccoli, cabbage

Supplements:
Phytomenadione (Vitamin K) tablets or injections

#END#

Minerals: Functions, Deficiencies, and Remedies

Minerals are inorganic elements required by the body for structural support, fluid balance, nerve signalling, enzyme activation, and other vital functions. They are divided into **macro-minerals** (needed in large amounts) and **trace minerals** (needed in small amounts but still essential).

#END#

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1. Calcium

Importance:

- Builds and maintains strong bones and teeth
- Required for muscle contraction and nerve impulse transmission
- Plays a role in blood clotting

Deficiency Diseases:

- **Rickets** (in children) – soft, weak bones
- **Osteomalacia/Osteoporosis** (in adults) – fragile bones

Symptoms:

- Bone pain and fractures
- Muscle cramps and spasms
- Weakness and numbness in extremities

Daily Requirement (RDA):

- Adults: 1000 mg/day

- Children: 700–1000 mg/day

Risk Factors:

- Low dairy intake
- Vitamin D deficiency
- Menopause (increased calcium loss)

Natural Sources:
Milk, cheese, yogurt, leafy greens, almonds, tofu, fortified cereals

Supplements:
Calcium carbonate or calcium citrate tablets

#END#

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2. Iron

Importance:

- Component of haemoglobin for oxygen transport in blood
- Supports metabolism and enzyme functions

Deficiency Disease:

- **Iron-Deficiency Anaemia**

Symptoms:

- Fatigue, pale skin
- Shortness of breath
- Brittle nails, hair loss

Daily Requirement:

- Men: 8 mg/day
- Women: 18 mg/day (higher in pregnancy)

Risk Factors:

- Vegetarian/vegan diets
- Heavy menstrual bleeding
- Pregnancy and lactation

Natural Sources:
Red meat, liver, spinach, lentils, beans, fortified cereals

Supplements:
Ferrous sulphate, ferrous fumarate tablets

#END#

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3. Magnesium

Importance:

- Muscle and nerve function
- Energy production and protein synthesis
- Maintains normal heart rhythm

Deficiency Disease:

- **Hypomagnesemia**

Symptoms:

- Muscle cramps, tremors
- Irregular heartbeat
- Fatigue, anxiety

Daily Requirement:

- Adults: 310–420 mg/day

Risk Factors:

- Alcoholism
- Gastrointestinal disorders (diarrhoea, Crohn's disease)
- Type 2 diabetes

Natural Sources:
Pumpkin seeds, spinach, almonds, black beans, avocado

Supplements:
Magnesium oxide or magnesium citrate

#END#

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4. Potassium

Importance:

- Maintains fluid and electrolyte balance
- Controls muscle contractions and nerve signals
- Helps regulate blood pressure

Deficiency Disease:

- **Hypokalaemia**

Symptoms:

- Muscle weakness and cramps
- Constipation

- Irregular heartbeat

Daily Requirement:

- Adults: 2600–3400 mg/day

Risk Factors:

- Excessive sweating
- Chronic diarrhoea or vomiting
- Diuretic medications

Natural Sources:
Bananas, oranges, potatoes, spinach, tomatoes

Supplements:
Potassium chloride (under medical supervision)

#END#

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5. Sodium

Importance:

- Regulates blood pressure and fluid balance
- Supports nerve signalling and muscle function

Deficiency Disease:

- **Hyponatremia**

Symptoms:

- Headache, confusion
- Seizures in severe cases
- Fatigue and low blood pressure

Daily Requirement:

- Adults: 1500–2300 mg/day (limit excess)

Risk Factors:

- Extreme sweating without replenishment
- Kidney disorders
- Excessive water intake

Natural Sources:
Salt, pickles, processed foods, seafood

Supplements:
Oral rehydration salts (ORS), sodium chloride tablets

#END#

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6. Zinc

Importance:

- Supports immune system
- Helps in wound healing and DNA synthesis
- Aids growth and development

Deficiency Disease:

- **Growth retardation**, delayed wound healing

Symptoms:

- Hair loss
- Loss of appetite
- Delayed puberty

Daily Requirement:

- Men: 11 mg/day
- Women: 8 mg/day

Risk Factors:

- Vegetarian diet without proper planning
- Chronic illnesses
- Alcohol abuse

Natural Sources:
Pumpkin seeds, oysters, cashews, beans

Supplements:
Zinc sulphate or zinc gluconate tablets

#END#

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7. Iodine

Importance:

- Essential for thyroid hormone production
- Controls metabolism and growth

Deficiency Disease:

- **Goitre**, hypothyroidism

Symptoms:

- Swelling in the neck

- Fatigue, weight gain
- Cognitive impairment in severe cases

Daily Requirement:

- Adults: 150 µg/day

Risk Factors:

- Diet low in seafood
- Living in iodine-deficient soil areas

Natural Sources:
Iodized salt, seaweed, fish, dairy products

Supplements:
Potassium iodide tablets

#END#

Other Essential Minerals

- **Selenium:** Antioxidant, thyroid support (Sources: Brazil nuts, eggs, tuna)
- **Phosphorus:** Bone health and energy metabolism (Sources: meat, dairy, nuts)
- **Copper:** Helps iron metabolism (Sources: shellfish, nuts, seeds)
- **Manganese:** Bone formation, enzyme activation (Sources: whole grains, nuts)
- **Fluoride:** Dental health (Sources: fluoridated water, tea)
- **Chromium:** Improves insulin action (Sources: broccoli, grapes)
- **Molybdenum:** Helps in amino acid metabolism (Sources: legumes, grains)

#END#

Disease/Condition: Ear Pain (Otalgia – non-severe, e.g., viral or simple otitis externa/media)
Overview: Ear pain can be due to infections (otitis media/externa), wax impaction, barotrauma, dental/sinus causes. Most viral/early cases are self-limiting; bacterial otitis media may need antibiotics. Seek urgent care if severe pain, high fever, discharge, dizziness, or hearing loss.
Common Symptoms: Earache, muffled hearing, fullness, sometimes fever, discharge, irritability in children.
Precautions: Do not insert objects into the ear; avoid water entry until cause clarified; seek care if worsening or systemic illness.
Prevention: Avoid Q-tips/deep cleaning; manage allergies/URIs; pneumococcal/flu vaccines for recurrent otitis.
Medications_and_Dosage:
– Paracetamol/acetaminophen: adults 500–1000 mg every 4–6 h (max 3 g/day preferred); children 10–15 mg/kg every 4–6 h (max per 24 h by weight).
– Ibuprofen: adults 200–400 mg every 6–8 h; children 5–10 mg/kg every 6–8 h.

– Topical analgesic ear drops (benzocaine/glycerin) may soothe, but avoid if tympanic membrane perforated.
– Antibiotics (e.g., amoxicillin) only if bacterial infection confirmed/prescribed.

#END#

Disease/Condition: Eye Pain / Red Eye (mild, non-vision threatening)
Overview: Causes include conjunctivitis, blepharitis, dry eyes, foreign body, or mild trauma. Dangerous causes include acute glaucoma, uveitis, corneal ulcer—urgent care needed if severe pain, vision loss, light sensitivity.
Common Symptoms: Eye redness, irritation, burning, tearing, mild discharge, eyelid swelling.
Precautions: Avoid rubbing; don’t share towels/cosmetics; remove contact lenses if symptoms occur; seek urgent care if severe.
Prevention: Good hygiene; avoid dirty hands in eyes; proper lens hygiene; eye protection at work/sports.
Medications_and_Dosage:
– Artificial tears (carboxymethylcellulose, hypromellose): 1–2 drops up to 4–6× daily (all ages).
– Antihistamine eye drops (olopatadine, ketotifen): adults/adolescents 1 drop twice daily; children ≥3 yrs per label.
– Paracetamol/ibuprofen as above for pain.
– Antibiotic eye drops (chloramphenicol, ofloxacin) only if bacterial conjunctivitis diagnosed by clinician.

#END#

Disease/Condition: Stomach Ache (nonspecific, mild, not surgical)
Overview: Common causes include indigestion, gastritis, constipation, mild viral gastroenteritis, food intolerance. Red flags: severe/sudden pain, persistent vomiting, blood in stool/vomit, weight loss—urgent evaluation.
Common Symptoms: Abdominal pain/discomfort, bloating, nausea, heartburn, constipation/diarrhea.
Precautions: Avoid self-medicating for severe pain; maintain hydration; seek care if persistent or severe.
Prevention: Balanced diet; avoid excessive NSAIDs/alcohol/spicy foods; hand hygiene; regular meals.
Medications_and_Dosage:
– Paracetamol for pain (avoid NSAIDs if suspected gastritis/ulcer).
– Antacids (aluminum hydroxide + magnesium hydroxide): adults 5–10 mL or 1–2 tablets as needed up to 4×/day; children/adolescents per label; infants—specialist only.
– Proton pump inhibitors (omeprazole 20 mg daily adults; pediatric dosing weight-based, specialist guided) for suspected reflux—short term.
– Oral rehydration solution for gastroenteritis—all ages.
– Laxatives (polyethylene glycol, lactulose) for constipation—adult/child dosing per label; avoid in infants without clinician.

#END#

Disease/Condition: Headache (tension-type, mild-moderate, not migraine/red-flag)
Overview: Most headaches are benign (tension, dehydration, fatigue, vision strain). Dangerous causes include meningitis, intracranial bleed, glaucoma—seek urgent care if sudden/severe or with neuro deficits.
Common Symptoms: Dull pressure, band-like pain, scalp/neck tightness, fatigue.
Precautions: Avoid excessive analgesic use (>10–15 days/month to prevent medication-overuse headache); check posture, hydration, sleep.
Prevention: Stress management, ergonomic posture, regular hydration, sleep hygiene.
Medications_and_Dosage:
– Paracetamol 500–1000 mg every 4–6 h (adults); children 10–15 mg/kg every 4–6 h.
– Ibuprofen 200–400 mg every 6–8 h (adults); children 5–10 mg/kg every 6–8 h.
– Avoid aspirin in children <16 years (Reye's syndrome risk).

#END#

Disease/Condition: Sore Throat (viral pharyngitis, not strep)
Overview: Often viral, self-limiting 3–7 days; bacterial (e.g., strep throat) requires antibiotics.
Red flags: drooling, airway difficulty, high fever, neck swelling.
Common Symptoms: Throat pain, difficulty swallowing, cough, mild fever, swollen tonsils.
Precautions: Rest, fluids, lozenges, avoid smoke; seek care if high fever or breathing/swallowing difficulty.
Prevention: Hand hygiene, avoid sharing utensils, mask if contagious.
Medications_and_Dosage:
– Paracetamol/ibuprofen for pain/fever (doses as above).
– Saltwater gargles, honey (avoid <1 yr), throat lozenges (older children/adults).
– Antibiotics (penicillin, amoxicillin) only if strep confirmed/prescribed.

#END#

Disease/Condition: Diarrhea (Acute, non-bloody, uncomplicated)
Overview: Commonly caused by viral gastroenteritis, contaminated food/water, or dietary intolerance. Usually self-limited (1–5 days). Risk of dehydration is highest in infants, elderly, and immunocompromised. Bloody diarrhea, high fever, or severe dehydration requires urgent evaluation (possible bacterial or parasitic cause).
Common Symptoms: Loose or watery stools, abdominal cramps, urgency, bloating, nausea, sometimes fever/vomiting.
Precautions: Maintain hydration; avoid antimotility drugs in children or with bloody diarrhea/fever; seek care for dehydration signs (dry mouth, sunken eyes, lethargy, very low urine output).
Prevention: Safe food/water hygiene; handwashing; rotavirus vaccination in infants; proper sanitation.
Medications_and_Dosage:
– Oral Rehydration Solution (ORS): cornerstone therapy, all ages; frequent small sips; children/infants—give after each loose stool (50–100 mL for <2 yrs; 100–200 mL for 2–10 yrs; as much as tolerated for older children/adults).
– Zinc supplementation (children 6 mo–5 yrs): 10–20 mg daily for 10–14 days (WHO guideline).
– Paracetamol for fever/pain (doses as above).
– Loperamide: adults only, 4 mg initially then 2 mg after each loose stool (max 16 mg/day);

not for children <12 yrs, or if bloody diarrhea/fever present.
– Antibiotics: only if bacterial etiology confirmed (e.g., azithromycin for traveler's diarrhea) per clinician.

#END#

Disease: Viral Fever

Overview:

- Viral fever is a nonspecific term used to describe fever caused by various viral infections, including adenovirus, enterovirus, influenza, dengue, or chikungunya.
- It is usually self-limiting, lasting 3–7 days depending on the causative virus.
- Patients may present with systemic symptoms like body ache, fatigue, and mild respiratory or gastrointestinal issues.
- Transmission occurs through droplets, contaminated food/water, or mosquito bites (for arboviral causes).
- Unlike bacterial infections, viral fevers generally do not respond to antibiotics.
- Management is mostly supportive with fluids, rest, and fever control.

Common Symptoms:

- Fever (low- to high-grade).
- Headache, body pain, muscle/joint pain.
- Sore throat, cough, or runny nose (in respiratory viruses).
- Fatigue, loss of appetite.
- In some cases, rash or mild diarrhea.

Precautions:

- Isolate patient during the fever period.
- Maintain good hydration and adequate rest.
- Avoid self-medication with antibiotics unless prescribed.

Prevention:

- Good hand hygiene.
- Use of mosquito nets/repellents in endemic areas.
- Seasonal influenza vaccination.
- Safe food and clean water practices.

Medications and Dosage:

- Paracetamol (Acetaminophen) for fever/pain:
  • Adults: 500–650 mg every 6–8 h (max 4 g/day).
  • Children: 10–15 mg/kg/dose every 6 h (max 60 mg/kg/day).

- Oral rehydration solutions and fluids.
- Antiviral drugs (e.g., Oseltamivir 75 mg twice daily for 5 days) only if influenza is confirmed.
- Avoid NSAIDs like aspirin in children (risk of Reye's syndrome).

#END#

Burns (Minor, First-Degree, or Small Second-Degree)

- Immediately place the affected area under cool (not ice-cold) running water for 15–20 minutes.
- Do **not** apply ice directly (can worsen tissue damage).
- Gently clean with mild soap and water.
- Ointments/Medications:
  • Aloe vera gel (soothing, natural healing).
  • Silver sulfadiazine cream (apply thin layer once/twice daily – avoid in children <2 months and solfa allergy).
  • Paracetamol: Adults 500–650 mg every 6–8 h; Children 10–15 mg/kg/dose every 6
- Cover loosely with a sterile, non-stick bandage.
- Avoid breaking blisters.
- **Consult a doctor if** burn is large, deep, involves the face/genitals, or shows signs of infection.

#END#

Cuts & Lacerations

- Wash hands before touching the wound.
- Rinse the cut with clean running water to remove dirt/debris.
- Use mild soap around the wound (not inside).
- Apply firm, gentle pressure with sterile gauze to stop bleeding.
- Once bleeding stops, apply an antiseptic solution (povidone-iodine) or antibiotic ointment.
- Cover with sterile bandage or dressing.
- Keep wound clean and change dressing daily.
- Ointments/Medications:
  • Povidone-iodine solution (antiseptic cleansing).
  • Neomycin or Mupirocin ointment (antibiotic protection, apply thin layer twice daily).
  • Paracetamol/Ibuprofen for pain relief.
- **Consult a doctor if** wound is deep, bleeding doesn't stop after 10 minutes, or signs of infection appear.

#END#

Abrasions (Skin Outs / Scrapes)

- Wash hands and gently clean the area with running water.
- Remove any dirt/foreign particles with sterile tweezers if needed.
- Apply antiseptic solution or antibiotic ointment.
- Cover with a clean, breathable bandage if the area is prone to dirt/rubbing.
- Leave minor scrapes uncovered at home to heal faster.
- Pain can be managed with Paracetamol.
- **Consult a doctor if** the wound is large, very dirty, or not healing in 3–5 days.

#END#

Bruises (Soft Tissue Injury without Cut)

- Apply a cold compress/ice pack wrapped in cloth for 10–15 minutes every 2–3 hours in the first 24 hours.
- Elevate the area if possible to reduce swelling.
- Avoid massaging the bruise.
- After 24–48 hours, use warm compresses to improve blood flow and healing.
- Pain relief: Paracetamol or Ibuprofen (Adults: 400 mg every 8 h, with food).
- **Consult a doctor if** bruise is very large, extremely painful, or associated with suspected fracture.

#END#

Minor Deformities / Sprains / Strains

- Follow **R.I.C.E.** method:
  R – Rest the injured area.
  I – Apply Ice pack (15–20 minutes every 2–3 h for first 24–48 h).
  C – Compression with elastic bandage (not too tight).
  E – Elevation above heart level to reduce swelling.
- Avoid putting weight on the affected joint.
- Pain relief: Ibuprofen 400 mg every 8 h (adults). Children: 10 mg/kg/dose every 6–8 h.
- **Consult a doctor if** swelling is severe, there’s deformity, inability to move joint, or fracture is suspected.

#END#